Implementing Edwards's Algorithm for Computing the Veech Group of a Translation Surface

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For Dad (1966–2018)

Abstract

A translation surface (X, ω) is a collection of polygons in $\mathbb{R}^2$ together with a set of side-pairing identifications such that identified sides are parallel, equal length and of opposite orientation. There is a natural action of $\mathrm{GL}_2\mathbb{R}$ on the set of all translation surfaces. A matrix M in the stabilizer of (X, ω) under the action of $\mathrm{GL}_2\mathbb{R}$ must preserve the area of (X, ω) and thus has determinant ± 1. The subgroup of orientation-preserving elements of this stabilizer is called the Veech group of (X, ω), denoted $\mathrm{SL}(X, \omega)$.

We present an implementation into SageMath of Edwards's algorithm for computing the Veech group of a compact translation surface (X, ω) ([Sa],[Ed]). Our algorithm builds on the sage-flatsurf package of SageMath developed by Delecroix and Hooper ([DeHo]). In general, the algorithm returns every element of $\mathrm{SL}(X, \omega)$ whose Frobenius norm is bounded above by a given constant. If the Veech group is a lattice, the algorithm returns a finite set of generators for $\mathrm{SL}(X, \omega)$ in finite time.

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1 Introduction and Background Material

In this section we provide an overview of the mathematical background necessary for understanding the Edwards algorithm presented in Section 2. This includes definitions and results pertaining to translation surfaces, the Veech group and hyperbolic geometry. Most of the results in this section are stated without proof, though resources for more detailed study are given throughout. We begin with a survey of other algorithms for computing Veech groups.

1.1 Other Algorithms

Algorithms for computing the Veech group of a translation surface began appearing in the literature in the early 2000s. Schmithüsen in [Sch] presents an algorithm for the restricted case that the translation surface is built of finitely many unit squares. Such a translation surface is called an origami, and it is known that the Veech group of an origami is a finite index subgroup of the modular group $\mathrm{SL}_2\mathbb{Z}$. As the modular group is generated by

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

all elements of the Veech group of an origami may be considered as words in S, T and their inverses. Her algorithm performs a breadth-first search of such words and tests Veech group membership of each until a set of generators is found. Finster in [Fre] extends this algorithm to the case that the translation surface is a cover of n-gons or double n-gons.

Both Bowman and Mukamel have developed algorithms for computing the Veech group of a general translation surface ([Bo],[Muk]). These algorithms produce candidate elements of the Veech group by studying the stabilizer of a cellular decomposition of the hyperbolic plane, then check whether or not each candidate belongs to the Veech group via the Delaunay triangulation of the translation surface.

In [SmWe], Smillie and Weiss prove that the Veech group of a translation surface is a lattice if and only if the surface has no small triangles. The authors mention that their work gives an implicit algorithm to find lattice surfaces, yet they conjecture that the algorithm is not 'practically feasible.' An exploration of this algorithm and improvements to it are offered in [Wu].

Another characterization of the Veech group is given in [BrJu], but an explicit method for computation is not provided. Their characterization of

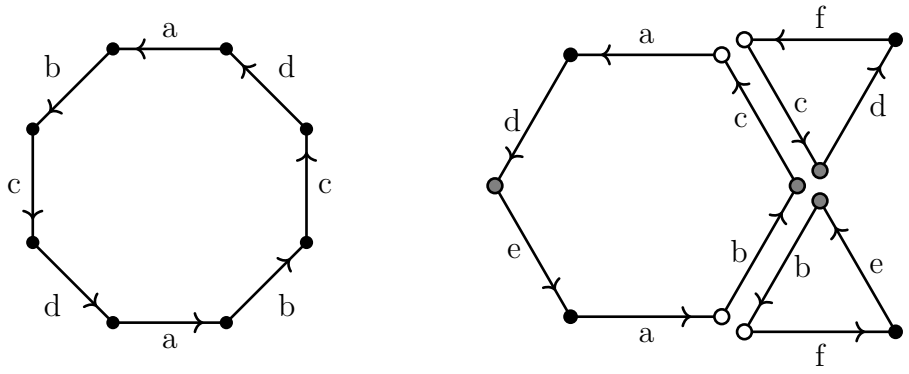


Figure 1: Left: The regular octagon forms a translation surface with one vertex of cone angle 6π. Right: A regular hexagon and two equilateral triangles form a translation surface with three vertices, each of cone angle 2π.

the Veech group is given in terms of orientation-preserving automorphisms of polygonal complexes arising from immersions of ellipses into translation surfaces.

The approach of the Edwards algorithm differs from those described above. For a given translation surface, the algorithm produces a model surface admitting an $SL_2\mathbb{R}$-action corresponding directly to the $SL_2\mathbb{R}$-action on the set of translation surfaces. Increasing subsets of the Veech group are found using this model surface, and these subsets give rise to hyperbolic polygons which nest down to a fundamental region for the action of the Veech group on the upper half-plane. If the Veech group is a lattice, the algorithm detects when such a hyperbolic polygon is in fact a fundamental region for the group, and a finite set of generators is returned.

1.2 Translation Surfaces

Here we give three equivalent definitions of a translation surface, each of which offers a unique and enlightening perspective in various situations. These definitions are discussed in detail in [Mas] and [Wri] where proofs of their equivalence may also be found.

The first definition is the most geometric and concrete of the three.

Definition 1.1. A *(polygonally presented) translation surface* $(\mathcal{P}, \mathcal{I})$ is a finite collection of polygons $\mathcal{P}$ in $\mathbb{R}^2$ together with a set of side-pairing iden-

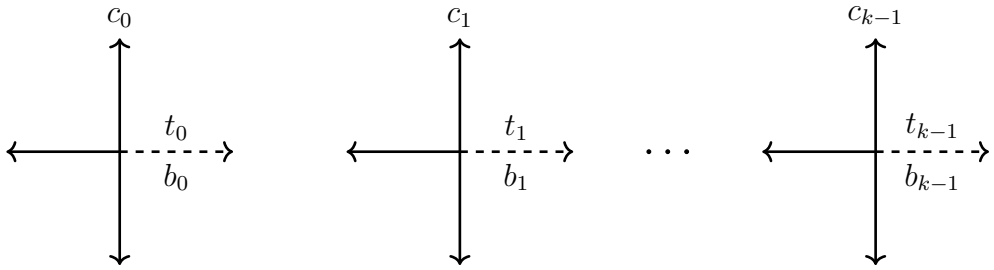


Figure 2: The surface $\mathcal{C}_k$, where the bottom of the slit, b_j, is identified with the top of the slit, t_{j+1}, for each j (modulo k).

tifications $\mathcal{I}$ such that each side of a polygon of $\mathcal{P}$ is identified with one other parallel, equal length and oppositely oriented side of a polygon of $\mathcal{P}$ so that the resulting surface is closed and connected.

From this definition, one finds that each non-vertex point p of a translation surface is contained in an open neighborhood that is homeomorphic to an open ball in $\mathbb{R}^2$. If p is a vertex, then the total angle about p is $2\pi k$ for some $k \in \mathbb{N}$. We call this angle the *cone angle* of the vertex p. For example, the octagon in Figure 1 has—after identifications are made—a single vertex of cone angle 6π. The hexagon and two triangles has three distinct vertices, each of cone angle 2π. We shall see these vertices arise in different guises in the two remaining definitions of a translation surface.

For the second definition, we begin by building a model surface by cyclically gluing together multiple copies of $\mathbb{C}$ in such a way that the final copy is glued to the initial copy (see Figure 2).

Definition 1.2. For each $k \in \mathbb{N}$, let $\mathcal{C}_k$ be an ordered set of k copies of $\mathbb{C}$, denoted $c_0, c_1, \ldots, c_{k-1}$, each of which is cut along the non-negative part of its real axis, and such that the lower edge of the resulting slit of c_j is identified with upper edge of the slit of c_{j+1} (where indexing is modulo k). The point $0 \in \mathcal{C}_k$ is the common origin of $c_0, \ldots, c_{k-1}$ after identifications are made.

An atlas on a real surface X is a collection of charts $\{(U_\alpha, \phi_\alpha)\}_{\alpha \in \mathcal{A}}$ such that $\{U_\alpha\}_{\alpha \in \mathcal{A}}$ forms an open cover for X and for each $\alpha \in \mathcal{A}$, the map $\phi_\alpha : U_\alpha \to \mathbb{R}^2$ is a homeomorphism onto its image (see [Lee] for a more detailed discussion).

Remark. We often identify $\mathbb{R}^2$ with $\mathbb{C}$ and move fluidly between these two

sets. Notation will clarify which perspective is used when explicit calculations are involved.

Given two charts $(U, \phi), (V, \psi)$ of an atlas with $U \cap V \neq \varnothing$, the transition function from U to V is the map

$$\psi \circ \phi^{-1} : \phi(U \cap V) \to \psi(V).$$

Definition 1.3. A *translation surface* is a closed and connected real surface X of genus $g \geq 1$ together with a finite subset $\Sigma \subset X$ such that $X \backslash \Sigma$ has a maximal atlas $\{(U_\alpha, \phi_\alpha)\}_{\alpha \in \mathcal{A}}$ whose transition maps are translations and such that each point $Q \in \Sigma$ is contained in an open neighborhood which is homeomorphic to $\mathcal{C}_k$ for some $k \in \mathbb{N}$. We call Q a *singularity of cone angle* $2\pi k$ and $\{(U_\alpha, \phi_\alpha)\}_{\alpha \in \mathcal{A}}$ a *translation atlas* on X.

A singularity of cone angle $2\pi k$ in this definition corresponds to a vertex of cone angle $2\pi k$ on a polygonally presented translation surface. If $k = 1$, we call the singularity *removable*. Hence the hexagon and two triangles in Figure 1 has three removable singularities.

The final definition of a translation surface requires the most mathematical background of the three and helps push forward the theory of translation surfaces where, for instance, the polygonally presented model would be tedious or inadequate to use. Conversely, concrete examples of translation surfaces are often given and studied as polygonally presented surfaces as they are more easily described and understood explicitly. We refer the reader to [FaKra] for more on the notions required for this final definition.

Definition 1.4. A *Riemann surface* X is a connected real surface together with a maximal atlas $\{(U_\alpha, \phi_\alpha)\}_{\alpha \in \mathcal{A}}$ of charts (into $\mathbb{C}$) whose transition functions are holomorphic. A *holomorphic* 1*-form* (also called an *abelian differential*) ω on a Riemann surface X is an assignment to each local coordinate on X an expression of the form

$$f(z)dz,$$

where f is holomorphic, and such that these assignments are invariant under coordinate changes. A *translation surface* (X, ω) is a closed Riemann surface X together with a non-zero holomorphic 1-form ω.

Given a translation surface (X, ω), a point $p \in X$ and a local coordinate z centered at p, we may write ω as $f(z)dz$ in this local coordinate by definition.

The *order of* ω *at* p is defined to be the order of vanishing of $f(z)$ at $z = 0$, which is independent of choice of local coordinate. If the order of ω at p is $k > 0$, then p is called a *zero of* ω *of order* k. If p is not a zero of ω, then there is a local coordinate z on X such that $\omega = dz$, and if p is a zero of order $k > 0$, then there is a local coordinate z on X such that $\omega = z^k dz$ (see Proposition 1.3 of [Wri]).

The zeros of ω correspond to the (non-removable) singularities in Σ of a translation surface given in Definition 1.3 and thus to vertices of cone angle greater than 2π of a translation surface given in Definition 1.1. If p is a zero of order k, then p is a singularity of cone angle $2\pi(k + 1)$ and corresponds to a vertex of cone angle $2\pi(k + 1)$ on a polygonally presented translation surface.

Because of the equivalence of these three definitions of a translation surface, we simply adopt the notation of the third definition and refer to a general translation surface as (X, ω).

1.3 The Veech Group

Following the presentation of [Ed], here we give two equivalent definitions of the Veech group of a translation surface (X, ω). We begin by defining the Veech group in terms of affine homeomorphisms of (X, ω). We then introduce an action of $\mathrm{GL}_2\mathbb{R}$ on the set of all translation surfaces and consider an equivalent definition of the Veech group that arises from the stabilizer of (an equivalence class of) (X, ω) under this action. The Edwards algorithm is best understood with both of these perspectives of the Veech group in mind.

Definition 1.5. Given two translation surfaces (X_1, ω_1) and (X_2, ω_2) with singular sets Σ_1 and Σ_2, respectively, a map

$$F : (X_1, \omega_1) \to (X_2, \omega_2)$$

sending Σ_1 into Σ_2 is called *affine* if it restricts to an affine map in the local coordinates on $X_1 \backslash \Sigma_1$. That is, given charts (U, ϕ) and (V, ψ) of the translation atlases on X_1 and X_2, respectively, with $F(U) \subseteq V$, we have that

$$\hat{F} := \psi \circ F \circ \phi^{-1} : \phi(U) \to \psi(V)$$

is of the form

$$\hat{F}(v) = Mv + b$$

for any $v \in \phi(U) \subseteq \mathbb{R}^2$, where $M \in M_2(\mathbb{R})$ is a real 2×2 matrix and $b \in \mathbb{R}^2$.

The translation structures and connectedness of X_1 and X_2 ensure that the matrix M in the above definition is globally constant for an affine map F. However, the translation b depends on local coordinates. That is, with notation as in Definition 1.5, if $(\tilde{U}, \tilde{\phi})$ and $(\tilde{V}, \tilde{\psi})$ are also charts of the same translation atlases on X_1 and X_2, respectively, with $F(\tilde{U}) \subseteq \tilde{V}$, then

$$\hat{\tilde{F}} : \tilde{\psi} \circ F \circ \tilde{\phi}^{-1} : \tilde{\phi}(\tilde{U}) \to \tilde{\psi}(\tilde{V})$$

is of the form

$$\hat{\tilde{F}}(v) = Mv + \tilde{b}$$

for some $\tilde{b} \in \mathbb{R}^2$. This matrix M is the Jacobian of the (locally represented) affine maps corresponding to F, which we denote by dF. The above definition of an affine map is, in general, a map between two distinct translation surfaces. The Veech group arises from self maps of this form.

Definition 1.6. Let (X, ω) be a translation surface and

$$\mathrm{Aff}^+(X, \omega) := \{F : (X, \omega) \to (X, \omega) \mid F \text{ is an orientation-preserving affine homeomorphism}\}.$$

The *Veech group* of (X, ω) is defined by

$$\mathrm{SL}(X, \omega) := \{dF \mid F \in \mathrm{Aff}^+(X, \omega)\}.$$

Notice that $\mathrm{SL}(X, \omega)$ is *a priori* simply a subset of the real 2×2 matrices $M_2(\mathbb{R})$. It is in fact a subgroup of $\mathrm{SL}_2\mathbb{R}$. The inverse of an orientation-preserving affine homeomorphism F is an orientation-preserving affine homeomorphism with $d(F^{-1}) = (dF)^{-1}$, so $\mathrm{SL}(X, \omega) \subseteq \mathrm{GL}_2\mathbb{R}$. For $F, G \in \mathrm{Aff}^+(X, \omega)$ one may check that $F \circ G \in \mathrm{Aff}^+(X, \omega)$ with $d(F \circ G) = (dF)(dG)$ so that $\mathrm{SL}(X, \omega) \leq \mathrm{GL}_2\mathbb{R}$. Furthermore, the fact that $F \in \mathrm{Aff}^+(X, \omega)$ is a homeomorphism from (X, ω) to itself implies that the area of (X, ω) is preserved and thus $\det(dF) = \pm 1$, while the orientation-preserving requirement of F gives $\det(dF) = 1$. That is, $\mathrm{SL}(X, \omega) \leq \mathrm{SL}_2\mathbb{R}$; hence the notation for the Veech group.

The notion of affine maps between translation surfaces gives rise to an equivalence relation on the set of all such surfaces.

Definition 1.7. Two translation surfaces (X_1, ω_1) and (X_2, ω_2) are called *translation equivalent* if there exists some affine map $F : (X_1, \omega_1) \to (X_2, \omega_2)$

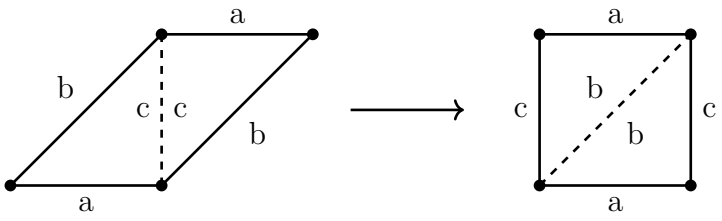


Figure 3: A cut along c and a paste along b give two translation equivalent translation surfaces.

with $dF = I_2$. We denote translation equivalence by $(X_1, \omega_1) \sim (X_2, \omega_2)$ and let

$$[(X_1, \omega_1)] := \{(X_2, \omega_2) \mid (X_1, \omega_1) \sim (X_2, \omega_2)\}.$$

To see that $\sim$ is in fact an equivalence relation, first note that the identity map is affine so that $(X_1, \omega_1) \sim (X_1, \omega_1)$. Affine maps are invertible with $d(F^{-1}) = (dF)^{-1}$, so $(X_1, \omega_1) \sim (X_2, \omega_2)$ implies $(X_2, \omega_2) \sim (X_1, \omega_1)$. Lastly, if F and G are affine maps whereby $(X_1, \omega_1) \sim (X_2, \omega_2)$ and $(X_2, \omega_2) \sim (X_3, \omega_3)$, respectively, then the composition

$$G \circ F : (X_1, \omega_1) \to (X_3, \omega_3)$$

is an affine map with $d(G \circ F) = (dG)(dF) = I_2$, so $(X_1, \omega_1) \sim (X_3, \omega_3)$.

Translation equivalence may be thought of more concretely in the polygonal setting as a 'cut-and-paste' equivalence as follows (see Figure 3): The polygons $\mathcal{P}$ of a translation surface (X, ω) may be 'cut' along a straight line whose endpoints are on the boundary and whose interior lies in the interior of a polygon of $\mathcal{P}$. A cut forms a new collection of polygons with side-pairing identifications inherited from $\mathcal{P}$ and with the new sides that arise from the cut also identified. We may also 'paste' together two identified sides of polygons of $\mathcal{P}$ by translating the polygons so that the identified edges coincide in $\mathbb{R}^2$. If a translation surface (X_2, ω_2) may be obtained after a finite sequence of cut-and-paste operations performed on (X_1, ω_1), then $(X_1, \omega_1) \sim (X_2, \omega_2)$.

If (X_1, ω_1) and (X_2, ω_2) are translation equivalent, then they have the same number of singularities of any particular cone angle. We say that $[(X, \omega)]$ has *orders* $i_1, \ldots, i_s \in \mathbb{N} \cup \{0\}$ if each representative $(X, \omega) \in [(X, \omega)]$ has s singularities of cone angles $2\pi(i_1 + 1), \ldots, 2\pi(i_s + 1)$. It is often convenient to consider the collection of all equivalence classes of translation surfaces with the same orders.

Definition 1.8. Let $i_1, \ldots, i_s \in \mathbb{N} \cup \{0\}$ with $i_j \leq i_{j+1}$ for $j = 1, \ldots, s-1$, and let

$$\mathcal{H}(i_1, \ldots, i_s) := \{[(X, \omega)] \mid [(X, \omega)] \text{ has orders } i_1, \ldots, i_s\}.$$

If $[(X, \omega)] \in \mathcal{H}(i_1, \ldots, i_s)$, we call $\mathcal{H}(i_1, \ldots, i_s)$ the *stratum* of (X, ω).

We next introduce an action of $\mathrm{GL}_2\mathbb{R}$ on the set of all translation surfaces and use this action to give an equivalent definition of the Veech group of a translation surface.

Definition 1.9. Let (X, ω) be a translation surface with singular set Σ and translation atlas $\{(U_\alpha, \phi_\alpha)\}_{\alpha \in \mathcal{A}}$, and let $M \in \mathrm{GL}_2\mathbb{R}$. We define $M \cdot (X, \omega)$ to be the translation surface given by the real surface X with singular set Σ and the translation atlas that arises from post-composing chart maps of $\{(U_\alpha, \phi_\alpha)\}_{\alpha \in \mathcal{A}}$ with M. That is, if T_M is the linear map (properly restricted) on $\mathbb{R}^2$ corresponding to the matrix M, then the translation atlas on $M \cdot (X, \omega)$ is $\{(U_\alpha, T_M \circ \phi_\alpha)\}_{\alpha \in \mathcal{A}}$.

One may check that $(M, (X, \omega)) \mapsto M \cdot (X, \omega)$ gives a well-defined group action of $\mathrm{GL}_2\mathbb{R}$ on the set of all translation surfaces. Furthermore, if $M \in \mathrm{GL}_2\mathbb{R}$ and (X, ω) is a translation surface,

$$M \cdot [(X, \omega)] := [M \cdot (X, \omega)]$$

gives a well-defined group action on the set of all equivalence classes of translation equivalent translation surfaces. Restricting this action to the subgroup $\mathrm{SL}_2\mathbb{R}$ leads to our second definition of the Veech group:

Definition 1.10. The Veech group of a translation surface is

$$\mathrm{SL}(X, \omega) := \mathrm{Stab}_{\mathrm{SL}_2\mathbb{R}}[(X, \omega)].$$

We refer the reader to the proof of Lemma 1 of [Ed] for the equivalence of Definitions 1.6 and 1.10.

1.4 Fuchsian Groups and Hyperbolic Geometry

In this subsection, we shift our setting to the upper half-plane

$$\mathbb{H} = \{z \in \mathbb{C} \mid \mathrm{Im}(z) > 0\}$$

and consider isometries given by an action of $\mathrm{PSL}_2\mathbb{R} := \mathrm{SL}_2\mathbb{R}/\{\pm I_2\}$ on $\mathbb{H}$. We introduce the notion of Fuchsian groups in $\mathrm{PSL}_2\mathbb{R}$ and give an explicit construction, known as the Dirichlet polygon, of a fundamental region corresponding to a given Fuchsian group. While these notions may initially seem disconnected from the ideas discussed in Subsections 1.2 and 1.3, we will see in Subsection 2.2 how the Dirichlet polygon construction is essential in the Edwards algorithm for determining a set of generators for the Veech group of a (lattice) translation surface. For the interested reader, Section 1 of [Sch] provides an overview of further applications of the study of the quotient of $\mathbb{H}$ by the action of a Veech group, and, in particular, how this setting is useful in understanding Teichmüller curves. We refer the reader to [Kat] and [Leh] for further background on the notions of hyperbolic geometry, Fuchsian groups and fundamental regions that comprise this subsection.

Remark. At the risk of being informal, we denote an element of $\mathrm{PSL}_2\mathbb{R}$ in what follows by either of its coset representatives. That is, by $M \in \mathrm{PSL}_2\mathbb{R}$ we mean

$$\{M, -M\} \in \mathrm{SL}_2\mathbb{R}/\{\pm I_2\}.$$

Unless otherwise stated, for $S \subseteq \mathrm{SL}_2\mathbb{R}$, we let $\overline{S}$ denote the image of S under the natural projection from $\mathrm{SL}_2\mathbb{R}$ to $\mathrm{PSL}_2\mathbb{R}$.

Identifying the set of real 2×2 matrices $M_2(\mathbb{R})$ with $\mathbb{R}^4$ via

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \sim (a, b, c, d),$$

the Euclidean norm on $\mathbb{R}^4$ suggests a norm on $M_2(\mathbb{R})$.

Definition 1.11. Let $M = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in M_2(\mathbb{R})$. The Frobenius norm of M is defined by

$$||M|| := \sqrt{a^2 + b^2 + c^2 + d^2}.$$

A quick calculation reveals that $||M|| = \sqrt{\mathrm{tr}(MM^t)}$. The Frobenius norm restricts to give a norm on $\mathrm{SL}_2\mathbb{R}$, and it passes to $\mathrm{PSL}_2\mathbb{R}$ as both coset representatives of an element of this group have the same Frobenius norm—each entry of these representative matrices differs by a sign, but the Frobenius norm squares each entry.

Definition 1.12. A discrete subgroup of $\mathrm{PSL}_2\mathbb{R}$ is called a *Fuchsian group*.

Veech in Proposition 2.7 of [Vee] and Vorobets in Proposition 3.3 of [Vor] show with different arguments that $\mathrm{SL}(X,\omega) \leq \mathrm{SL}_2\mathbb{R}$ is a discrete group, from which it follows that the image of $\mathrm{SL}(X,\omega)$ in $\mathrm{PSL}_2\mathbb{R}$—which we denote $\mathrm{PSL}(X,\omega)$—is a Fuchsian group. There is an isometric action of Fuchsian groups on the upper half-plane $\mathbb{H}$, but in order to make sense of this statement we first need a metric on $\mathbb{H}$.

Let $\gamma : [0,1] \to \mathbb{H}$ be a piecewise differentiable curve parametrized by $\gamma(t) = x(t) + iy(t)$. The real-valued integral

$$h(\gamma) := \int_0^1 \frac{\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}}{y} dt$$

is called the *hyperbolic length* of γ. Fix $w, z \in \mathbb{H}$ and let $\mathcal{C}_w^z$ be the set of all piecewise differentiable curves in $\mathbb{H}$ beginning at w and ending at z. Define

$$\rho(w,z) := \inf_{\gamma \in \mathcal{C}_w^z} h(\gamma).$$

It can be shown that ρ defines a metric on $\mathbb{H}$, which we call *hyperbolic distance*, and that $\rho(w,z)$ is realized by a unique curve γ, which we call the *hyperbolic geodesic* from w to z (unique referring to the image of γ, as two distinct curves may have the same image in $\mathbb{H}$ though they are parameterized by different 'speeds'; see Corollary 1.2.2 of [Kat]).

There is also a notion of volume in $\mathbb{H}$. Given a subset $A \subseteq \mathbb{H}$, the *hyperbolic volume* of A is defined by

$$\nu(A) := \int_A \frac{dxdy}{y^2}$$

when the integral exists. We mention here that ν is a measure on $\mathbb{H}$. This follows from the Beppo Levi Theorem, which is equivalent to the Monotone Convergence Theorem (Corollary 18.11 of [Car]). In particular, ν is countably additive. While this volume can often be explicitly computed with calculus techniques, the following result simplifies many volume calculations (see Theorem 1.4.2 of [Kat]):

Lemma 1.1 (Gauss-Bonnet Theorem)**.** *Let T be a hyperbolic triangle (that is, a region in $\mathbb{H}$ bounded by three hyperbolic geodesics) with interior angles α, β and γ. Then*

$$\nu(T) = \pi - (\alpha + \beta + \gamma).$$

We call a subset A of the upper half-plane $\mathbb{H}$-*convex* if the hyperbolic geodesic connecting any two points of the set A is completely contained in A. An immediate consequence of the Gauss-Bonnet Theorem is the following:

Corollary 1.1. *Let $\mathcal{P} \subset \mathbb{H}$ be an $\mathbb{H}$-convex hyperbolic polygon with boundary $\partial\mathcal{P}$ comprised of finitely many hyperbolic geodesics, and let $\alpha_0, \ldots, \alpha_{n-1}$ be the interior angles of the vertices of $\mathcal{P}$. Then*

$$\nu(\mathcal{P}) = (n-2)\pi - \sum_{k=0}^{n-1} \alpha_k.$$

Proof. Since $\mathcal{P}$ is $\mathbb{H}$-convex, we may triangulate $\mathcal{P}$ by fixing some point τ in its interior and drawing a geodesic from τ to each of the vertices of $\mathcal{P}$ (see Figure 4). Relabeling if necessary, we may assume that $\alpha_0, \ldots, \alpha_{n-1}$ correspond to subsequent vertices $v_0, \ldots, v_{n-1}$ as one moves counterclockwise about $\partial\mathcal{P}$. Let $T_0, \ldots, T_{n-1}$ denote the triangles comprising the triangulation of $\mathcal{P}$ so that T_j has v_j, v_{j+1} and τ as its three vertices (with indexing here and below modulo n). Let $\beta_j, \gamma_j, \delta_j$ be the internal angles of the three vertices of T_j corresponding to v_j, v_{j+1}, and τ, respectively. Note that

$$\sum_{j=0}^{n-1} \delta_j = 2\pi \qquad \text{and} \qquad \gamma_{j-1} + \beta_j = \alpha_j$$

for each j. From the Gauss-Bonnet Theorem (Lemma 1.1), we find

$$\begin{aligned}
\nu(\mathcal{P}) &= \sum_{k=0}^{n-1} \nu(T_k) \\
&= \sum_{k=0}^{n-1} (\pi - (\beta_k + \gamma_k + \delta_k)) \\
&= n\pi - \left(\gamma_{n-1} + \beta_0 + \sum_{k=1}^{n-1} (\gamma_{k-1} + \beta_k) + \sum_{j=0}^{n-1} \delta_j \right) \\
&= n\pi - \left(\sum_{k=0}^{n-1} \alpha_k + 2\pi \right) \\
&= (n-2)\pi - \sum_{k=0}^{n-1} \alpha_k.
\end{aligned}$$

□

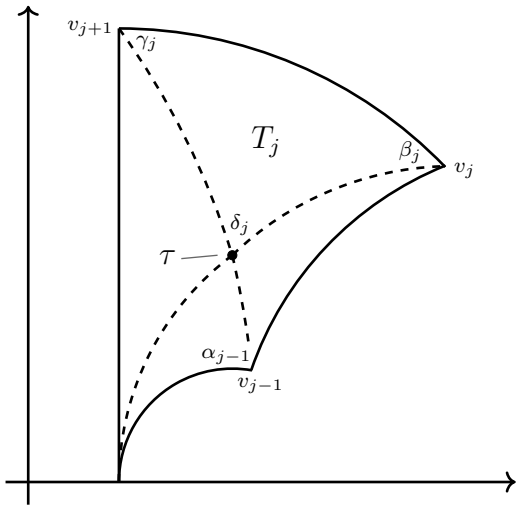


Figure 4: Triangulation of a hyperbolic polygon as in the proof of Corollary 1.1.

There is a well-studied group action of $\mathrm{PSL}_2\mathbb{R}$ by Möbius transformations on the upper half-plane $\mathbb{H}$:

$$\mathrm{PSL}_2\mathbb{R} \times \mathbb{H} \to \mathbb{H}$$
$$\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}, z \right) \mapsto \frac{az+b}{cz+d}.$$

When $\mathbb{H}$ is endowed with the hyperbolic metric introduced above, $\mathrm{PSL}_2\mathbb{R}$ acts on $\mathbb{H}$ by isometry (Theorem 1.1.2 of [Kat]). That is, for $M \in \mathrm{PSL}_2\mathbb{R}$ and $w, z \in \mathbb{H}$, we have

$$\rho(M \cdot w, M \cdot z) = \rho(w, z).$$

Similarly, the action of $\mathrm{PSL}_2\mathbb{R}$ preserves hyperbolic area so that $\nu(M \cdot A) = \nu(A)$ for $M \in \mathrm{PSL}_2\mathbb{R}$ and $A \subseteq \mathbb{H}$ whenever $\nu(A)$ exists (Theorem 1.4.1 of [Kat]).

In general, Möbius transformations (maps on $\mathbb{C} \cup \{\infty\}$ of the form $z \mapsto \frac{az+b}{cz+d}$ with $a, b, c, d \in \mathbb{C}, ad - bc \neq 0$) map circles and straight lines to circles and straight lines, and they are conformal in the sense that angles are preserved in the image. These properties hold when we restrict to the Möbius transformations arising from the action of $\mathrm{PSL}_2\mathbb{R}$ on $\mathbb{H}$. Note that if we extend this action to $\overline{\mathbb{H}} = \mathbb{H} \cup \mathbb{R} \cup \{\infty\}$, then $\partial\mathbb{H} = \mathbb{R} \cup \{\infty\}$ is sent to itself.

From this and the conformal property of Möbius transformations, it follows that the action of $\mathrm{PSL}_2\mathbb{R}$ on $\mathbb{H}$ sends the set of semicircles in $\mathbb{H}$ centered on $\mathbb{R}$ together with vertical lines to the same set of semicircles and vertical lines. We call this set of semicircles and vertical lines *h-lines*. The group $\mathrm{PSL}_2\mathbb{R}$ acts transitively on the set of h-lines (this can be seen by showing that every h-line is sent to the positive imaginary axis by some element of $\mathrm{PSL}_2\mathbb{R}$ and using the fact that every element of $\mathrm{PSL}_2\mathbb{R}$ is invertible).

A quick exercise using the definition of the hyperbolic metric ρ reveals that the unique geodesic realizing $\rho(ai, bi)$ for any $0 < a < b$ is the vertical line segment from ai to bi (see the proof of Theorem 1.2.1 of [Kat]). Fix $w, z \in \mathbb{H}$, and let ℓ be the unique h-line passing through both w and z. Let γ be the segment of ℓ beginning at w and ending at z. By the transitivity of $\mathrm{PSL}_2\mathbb{R}$ on the set of h-lines, there is some $M \in \mathrm{PSL}_2\mathbb{R}$ sending ℓ to the positive imaginary axis. Now $M \cdot \gamma$ is the unique geodesic realizing $\rho(M \cdot w, M \cdot z)$ as it is the vertical line segment from $M \cdot w$ to $M \cdot z$ on the imaginary axis. But M is an isometry, from which we conclude the following:

Lemma 1.2. *The set of geodesics in $\mathbb{H}$ is the set of connected segments of h-lines.*

We next define a fundamental region for the action of a Fuchsian group on $\mathbb{H}$ and consider a particular construction of a fundamental region, called the Dirichlet polygon, which carves out a subset of $\mathbb{H}$ that is bounded by geodesics determined by the Fuchsian group. We then present results about the Dirichlet polygon that are used in the Edwards algorithm.

Definition 1.13. Let $\mathcal{F}$ be the closure of a non-empty open set. We call $\mathcal{F}$ a *fundamental region* for the action of a Fuchsian group $\Gamma \leq \mathrm{PSL}_2\mathbb{R}$ on $\mathbb{H}$ if

(i) every point in $\mathbb{H}$ is in the Γ-orbit of a point in $\mathcal{F}$;

(ii) if $w, z \in \mathcal{F}$ are distinct points of the same Γ-orbit, then $w, z \in \partial\mathcal{F}$, and;

(iii) the boundary of $\mathcal{F}$ has zero hyperbolic volume.

If a fundamental region $\mathcal{F}$ for the action of a Fuchsian group Γ has finite hyperbolic volume, then we call Γ a *lattice* group. If $G \leq \mathrm{SL}_2\mathbb{R}$ such that $\overline{G} \leq \mathrm{PSL}_2\mathbb{R}$ is a lattice group, we also refer to G itself as a lattice group.

Remark. Most authors do not require condition (iii) of Definition 1.13. The fundamental regions we consider will have boundaries satisfying this condition; we include it in our definition so as to omit its explicit statement in the hypotheses of results throughout this subsection.

An extremely useful result about fundamental regions of lattice groups and their subgroups is the following:

Lemma 1.3 (See Theorem 3.1.2 of [Kat], whose proof also goes through for the infinite index case)**.** *Let $\Gamma \leq PSL_2\mathbb{R}$ be a lattice group and $\Lambda \leq \Gamma$ a (finite or infinite index) subgroup. If $\mathcal{F}_\Gamma$ and $\mathcal{F}_\Lambda$ are fundamental regions for the actions of Γ and Λ, respectively, then*

$$\nu(\mathcal{F}_\Lambda) = |\Gamma : \Lambda| \ \nu(\mathcal{F}_\Gamma).$$

A fundamental region for the action of a Fuchsian group on $\mathbb{H}$ is by no means unique (though Lemma 1.3 implies the hyperbolic volumes of two fundamental regions for a group are equal). In fact, given a fundamental region $\mathcal{F}$ for a group Γ, the collection of sets $\{M \cdot \mathcal{F}\}_{M\in\Gamma}$ form what is called a *tessellation* of $\mathbb{H}$, and each particular $M \cdot \mathcal{F}$ is a fundamental region for Γ. As we made no restriction on connectedness in the definition of a fundamental region, one can imagine 'stitching' together pieces of various sets comprising a tessellation $\{M \cdot \mathcal{F}\}_{M\in\Gamma}$ to form a new fundamental region for Γ that is distinct from each $M \cdot \mathcal{F}$ in the tessellation. This is all to say that fundamental regions for a particular Fuchsian group are numerous and can be quite complicated. However, the Dirichlet polygon defined below provides a concrete construction of a connected, $\mathbb{H}$-convex hyperbolic polygon giving a fundamental region for a Fuchsian group Γ. For this construction, we introduce notation for the closed half-plane that contains all points that are closer to a fixed $\tau \in \mathbb{H}$ than to a distinct point $z \in \mathbb{H}$:

Definition 1.14. Fix $\tau \in \mathbb{H}$. For any $z \in \mathbb{H}\backslash\{\tau\}$, define

$$H_\tau(z) := \{w \in \mathbb{H} \mid \rho(w, \tau) \leq \rho(w, z)\}.$$

The set $H_\tau(z)$ may be explicitly constructed as follows: let γ be the unique geodesic connecting τ and z, and let $\gamma^\perp$ be the hyperbolic perpendicular bisector of γ. That is, $\gamma^\perp$ is the h-line passing orthogonally through the midpoint of γ with respect to the hyperbolic distance ρ. The h-line $\gamma^\perp$ defines two open hyperbolic half-planes—one containing τ and one containing z. The set $H_\tau(z)$ is the closure of the hyperbolic half-plane containing τ.

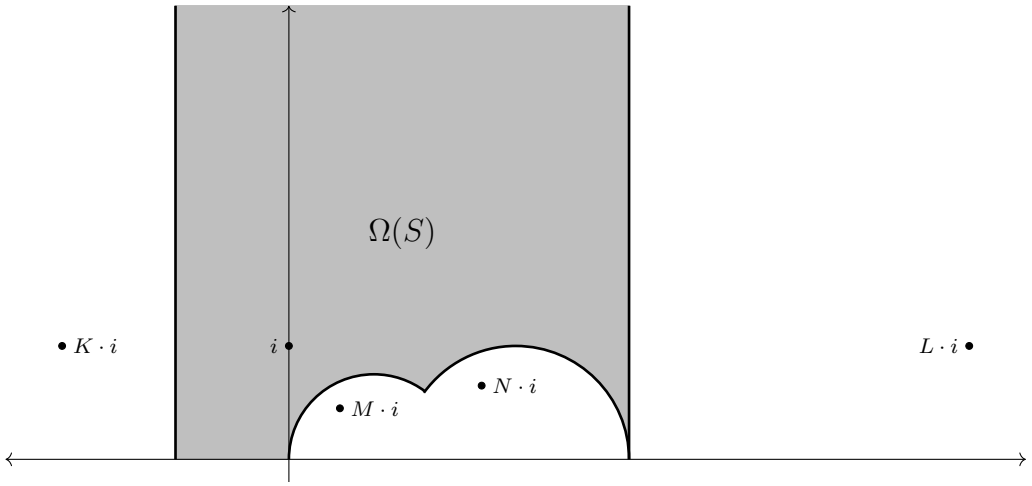


Figure 5: The hyperbolic polygon $\Omega(S)$ with $S = \{K, L, M, N\} \subset \mathrm{PSL}_2\mathbb{R}$ has one free side, one vertex in $\mathbb{H}$ and two vertices at infinity.

Definition 1.15. Let $\tau \in \mathbb{H}$ and $S \subset \mathrm{PSL}_2\mathbb{R}$ be discrete such that for any $M \in S$, $M \cdot \tau = \tau$ implies $M = \mathrm{Id}$. We define the *hyperbolic polygon determined by S and centered at τ* to be

$$\Omega_\tau(S) := \bigcap_{M \in S \setminus \{\mathrm{Id}\}} H_\tau(M \cdot \tau).$$

If Γ is a Fuchsian group that trivially stabilizes $\tau \in \mathbb{H}$, we call $\Omega_\tau(\Gamma)$ the *Dirichlet polygon of Γ centered at τ*.

A Dirichlet polygon for a Fuchsian group Γ always exists since almost every $\tau \in \mathbb{H}$ is fixed only by $\mathrm{Id} \in \Gamma$ (Lemma 2.2.5 of [Kat]). In practice, we will focus mainly on hyperbolic and Dirichlet polygons centered at $\tau = i$ and use the suppressed notation $\Omega(S) := \Omega_i(S)$. The stabilizer of i under the action of $\mathrm{PSL}_2\mathbb{R}$ on $\mathbb{H}$ is precisely the image of orientation-preserving orthogonal matrices, $\overline{\mathrm{SO}_2\mathbb{R}} \leq \mathrm{PSL}_2\mathbb{R}$. That is, $\mathrm{Stab}_{\mathrm{PSL}_2\mathbb{R}}(i)$ is the image in $\mathrm{PSL}_2\mathbb{R}$ of matrices that act as rigid rotations in $\mathbb{R}^2$. Elements of $\mathrm{SO}_2\mathbb{R}$ appear in the Veech group of a translation surface (X, ω) when (X, ω) has—loosely speaking—rotational symmetry. When this is the case, step 2 of the Edwards algorithm (Algorithm 2.1) perturbs (X, ω) by some matrix M_0 so that $\mathrm{PSL}(M_0 \cdot (X, \omega))$ trivially stabilizes i. We then study $\Omega(S)$ for increasing subsets $S \subset \mathrm{PSL}(M_0 \cdot (X, \omega))$ to find generators of $\mathrm{PSL}(M_0 \cdot (X, \omega))$, and the

desired group $\mathrm{PSL}(X,\omega)$ is obtained from $\mathrm{PSL}(M_0 \cdot (X,\omega))$ by conjugation under M_0^{-1} (step 11 of Algorithm 2.1).

Given a discrete subset $S \subset \mathrm{PSL}_2\mathbb{R}$ and $M \in S\backslash\{\mathrm{Id}\}$, each half-plane $H_i(M \cdot i)$ in the construction of the hyperbolic polygon determined by S is bounded by the perpendicular bisector $\gamma_M^{\perp}$ of the geodesic γ_M from i to $M \cdot i$. Each of these half-planes is $\mathbb{H}$-convex, and so their intersection $\Omega(S)$ is also $\mathbb{H}$-convex. The boundary of $\Omega(S)$ is comprised of segments of the various perpendicular bisectors $\gamma_M^{\perp}$, which we call *sides* of $\Omega(S)$, and segments of $\mathbb{R}$, which we call *free sides* of $\Omega(S)$. We call the intersection of two sides in $\mathbb{H}$ a *vertex* of $\Omega(S)$. If a side intersects a side in $\mathbb{R}$, we call this intersection a *vertex at infinity*. If two sides are segments of vertical h-lines with no vertices giving an upper bound to either of these segments, we also say that the two sides share a *vertex at infinity* (see Figure 5).

We state here some useful results about Dirichlet polygons:

Lemma 1.4. *Let $\Gamma \leq PSL_2\mathbb{R}$ be a Fuchsian group that trivially stabilizes $i \in \mathbb{H}$. Then*

(i) (Theorem 3.2.2 of [Kat]) $\Omega(\Gamma)$ is a connected fundamental region for the action of Γ on $\mathbb{H}$;

(ii) (Chapter 1 Section 5C of [Leh] and Corollary 1.1) $\nu(\Omega(\Gamma)) < \infty$ if and only if $\Omega(\Gamma)$ has finitely many sides, none of which are free sides;

(iii) (Section 3.5 of [Kat]) there is a subset $G \subset \Gamma$ containing elements (called the side-pairing transformations of $\Omega(\Gamma)$) such that each side of $\Omega(\Gamma)$ is sent by exactly one $M \in G$ to another side of $\Omega(\Gamma)$;

(iv) (Theorem 3.5.4 of [Kat] and Chapter 1 Section 4F of [Leh]) the set of side pairing transformations $G \subset \Gamma$ is the minimal subset of Γ such that $\Omega(G) = \Omega(\Gamma)$, and G forms a set of generators for Γ.

For statement (i) of Lemma 1.4, the first two conditions of a fundamental region (Definition 1.13) are verified by the aforementioned Theorem 3.2.2 of [Kat]. The third condition is satisfied since the boundary of a Dirichlet polygon is the union of at most countably many (from the discreteness of Fuchsian groups) geodesics and possibly segments of $\mathbb{R}$ and since ν is countably additive as a measure on $\mathbb{H}$. Note that statement (ii) of Lemma 1.4 gives necessary and sufficient conditions for $\mathrm{PSL}(X,\omega)$ to be a lattice group. In this case, the Edwards algorithm finds $\Omega(\mathrm{PSL}(X,\omega))$, and parts (iii) and (iv) of Lemma 1.4 provide generators for $\mathrm{PSL}(X,\omega)$.

2 The Algorithm

In this section we present the Edwards algorithm for computing the Veech group of a translation surface (X, ω). Subsections 2.1 and 2.2 provide an overview of the main tools that are constructed by Edwards for the algorithm itself. We refer the reader to various sections of [Ed] for more details regarding these constructions. The main object introduced in Subsection 2.1 is a model surface that admits a computable $\mathrm{SL}_2\mathbb{R}$-action and that records sufficient information about (X, ω) so that one may reconstruct (X, ω) from the model surface. We will see that this action on the model surface gives rise to another characterization of the Veech group of (X, ω) (Theorem 2.2) and that increasing sets of finite data stored in the model surface can be used to find increasing subsets of $\mathrm{SL}(X, \omega)$ (Theorem 2.3). In the case that $\mathrm{SL}(X, \omega)$ is a lattice, the algorithm uses the hyperbolic polygon construction introduced in Definition 1.15 to recognize when increasing subsets of $\mathrm{SL}(X, \omega)$ contain a full set of generators for the group (see Subsection 2.2). We give a recurrent example throughout Subsections 2.1 and 2.2 to highlight the key notions of the algorithm. Subsection 2.3 presents an improved version of the Edwards algorithm in pseudocode, and Subsection 2.4 highlights the changes that are made to the original algorithm with justifications of their validity.

2.1 A Model Surface for Finding Veech Group Elements

Recall that an affine map $F : (X_1, \omega_1) \to (X_2, \omega_2)$ sends the singular set Σ_1 of (X_1, ω_1) into the singular set Σ_2 of (X_2, ω_2) (Definition 1.5). Thus an orientation-preserving affine homeomorphism $F \in \mathrm{Aff}^+(X, \omega)$ sends the singular set Σ of (X, ω) into itself. This observation suggests that the study of the set Σ is paramount to finding elements of $\mathrm{SL}(X, \omega)$, as it puts a stringent restriction on possible self-maps $F \in \mathrm{Aff}^+(X, \omega)$.

Definition 2.1. Let (X, ω) be a translation surface with singular set Σ. A *saddle connection* is an oriented geodesic γ on (X, ω) with endpoints in Σ and interior lying in $X \backslash \Sigma$. Let $\mathrm{SC}(X, \omega)$ denote the set of all saddle connections on (X, ω) and $\mathrm{SC}_Q(X, \omega)$ the subset of saddle connections in $\mathrm{SC}(X, \omega)$ which

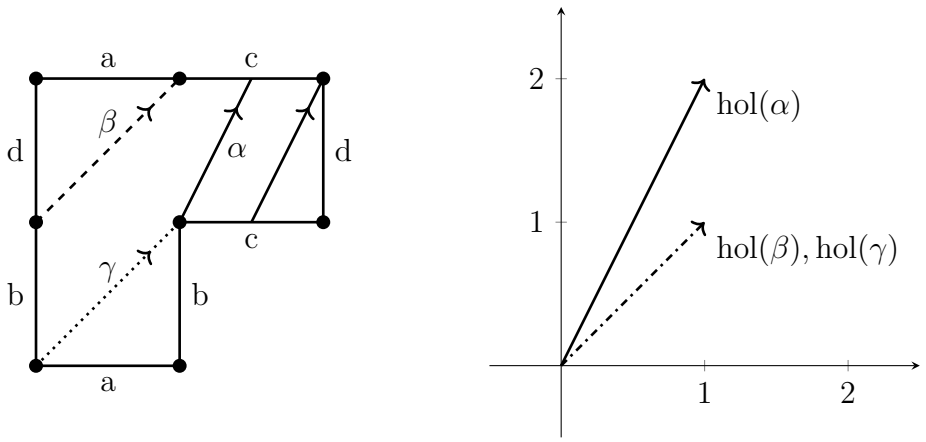


Figure 6: Left: Three distinct saddle connections on (R, ψ). Right: The saddle connections' corresponding holonomy vectors.

begin at $Q \in \Sigma$. We define the *holonomy map* by

$$\begin{aligned} \mathrm{hol} : \mathrm{SC}(X, \omega) &\to \mathbb{C} \\ \gamma &\mapsto \int_\gamma \omega \end{aligned}$$

and call $\mathrm{hol}(\gamma)$ the *holonomy vector* of the saddle connection $\gamma \in \mathrm{SC}(X, \omega)$.

Let (R, ψ) be the translation surface formed by gluing together three unit squares as seen in Figure 6. (We will return to (R, ψ) throughout this and the next subsection.) A saddle connection is a straight line beginning and ending at singularities of (R, ψ) that respects the side-pairing identifications of the surface. For example, the saddle connection α leaves the upper edge c and continues in the same direction from the identified point on the lower edge c. The corresponding holonomy vector is the vector of equal length and direction developed from the origin in the plane. This may be thought of as either a vector in $\mathbb{R}^2$ or a point in $\mathbb{C}$. Notice that the holonomy map is not injective, as $\mathrm{hol}(\beta) = \mathrm{hol}(\gamma)$ for $\beta \neq \gamma$.

Since an orientation-preserving affine homeomorphism $F \in \mathrm{Aff}^+(X, \omega)$ sends Σ to itself, it follows that F sends saddle connections to saddle connections on (X, ω), i.e.

$$F(\mathrm{SC}(X, \omega)) \subseteq \mathrm{SC}(X, \omega).$$

Invertibility of F implies that the reverse containment also holds, so

$$F(\mathrm{SC}(X,\omega)) = \mathrm{SC}(X,\omega).$$

In a sense, holonomy vectors describe saddle connections up to translations arising from edge identifications on (X,ω) just as the Jacobian $M = dF$ describes the affine homeomorphism F up to local translations. In fact, elements of the Veech group preserve the set of holonomy vectors of a translation surface. That is, if $M \in \mathrm{SL}(X,\omega)$, then

$$M(\mathrm{hol}(\mathrm{SC}(X,\omega))) = \mathrm{hol}(\mathrm{SC}(X,\omega)) \tag{1}$$

where $\mathrm{hol}(\mathrm{SC}(X,\omega))$ is considered as a subset of $\mathbb{R}^2$ and $M(\mathrm{hol}(\mathrm{SC}(X,\omega)))$ represents the image of $\mathrm{hol}(\mathrm{SC}(X,\omega))$ under matrix multiplication (see Section 3 of [Vor]; this is also a direct consequence of Theorem 16 of [Ed]). Hence $\mathbb{R}^2$ appears to be a natural setting to search for elements of $\mathrm{SL}(X,\omega)$: find the saddle connections on (X,ω), develop their holonomy vectors in $\mathbb{C} \sim \mathbb{R}^2$, and construct matrices in $\mathrm{SL}_2\mathbb{R}$ that preserve this set of holonomy vectors. However, as noted above, $\mathrm{hol}(\cdot)$ is not injective and so the image of $\mathrm{SC}(X,\omega)$ under $\mathrm{hol}(\cdot)$ loses information about the original surface (X,ω). In general, it is too easy for a matrix to preserve the set of holonomy vectors; a matrix $M \in \mathrm{SL}_2\mathbb{R}$ may satisfy equation (1) but not be an element of $\mathrm{SL}(X,\omega)$.

The idea of the Edwards algorithm is to construct a space $\mathcal{O}$ where an analogue of holonomy vectors may be developed in such a way that no information is lost as it is in $\mathbb{C}$. In fact, we will see that the space $\mathcal{O}$ stores enough information about (X,ω) that a finite set of data uniquely determines a translation surface up to translation equivalence (see Theorem 2.1). Furthermore, there is an action of $\mathrm{SL}_2\mathbb{R}$ on $\mathcal{O}$ corresponding to the action of $\mathrm{SL}_2\mathbb{R}$ on the set of equivalence classes of translation equivalent translation surfaces. In theory, the algorithm begins with an equivalence class $[(X,\omega)]$, pushes data about a representative (X,ω) to the space $\mathcal{O}$, finds elements of $\mathrm{SL}_2\mathbb{R}$ that preserve this data, then reconstructs a representative of $[(X,\omega)]$ so that the matrices that are found actually belong to $\mathrm{SL}(X,\omega)$ (see Figure 7). In practice, the final step of reconstructing (X,ω) is omitted, as we are just in search of $\mathrm{SL}(X,\omega)$ and the matrices that preserve the recorded data in $\mathcal{O}$ turn out to be exactly this group.

Definition 2.2 (See Definition 13 and Section 3.3 of [Ed])**.** Let (X,ω) be a translation surface with singular set Σ. Suppose (X,ω) belongs to the

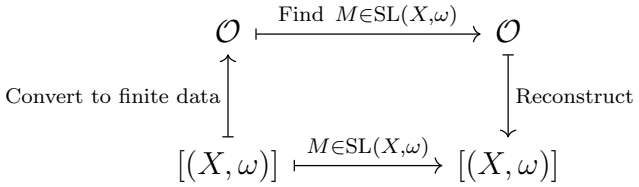


Figure 7: Using $\mathcal{O}$ to find elements of $\mathrm{SL}(X,\omega)$.

stratum $\mathcal{H}(i_1,\ldots,i_s)$ (see Definition 1.8), and let $n(k)$ denote the number of times $k \in \mathbb{N} \cup \{0\}$ occurs in $i_1,\ldots,i_s$. For each $k \in \mathbb{N} \cup \{0\}$, let $\mathcal{O}_k$ be the non-compact Riemann surface $\mathbb{C}$ together with the 1-form dz^{k+1}. We define

$$\mathcal{O} := \bigsqcup_{k \in \mathbb{N} \cup \{0\}} \left(\bigsqcup_{j=1}^{n(k)} \mathcal{O}_k \right).$$

Each $\mathcal{O}_k$ may be thought of as $k+1$ copies of $\mathbb{C}$ glued together cyclically (see Definition 1.2 for this approach, where $\mathcal{O}_k = \mathcal{C}_{k+1}$.) Observe that in the definition of $\mathcal{O}$, $n(k)$ is nonzero for only finitely may k, so this is a finite disjoint union. In fact, for each $Q \in \Sigma$ of order k—or cone angle $2\pi(k+1)$—there is a distinct model surface $\mathcal{O}_k$ in the disjoint union $\mathcal{O}$. If we let $\mathcal{O}_Q$ represent a copy of $\mathcal{O}_k$, then we may identify $\sqcup_{Q \in \Sigma} \mathcal{O}_Q$ with $\mathcal{O}$, but there is no canonical identification. Distinct identifications differ by permutations of the $n(k)$ copies of $\mathcal{O}_k$ for each $k \in \mathbb{N} \cup \{0\}$.

Whereas the map sending saddle connections on (X,ω) to holonomies in $\mathbb{C}$ is not injective, the surface $\mathcal{O}$ is constructed to provide enough 'space' so that saddle connections may be developed injectively. We call these vectors in $\mathcal{O}$ *marked periods* of (X,ω) to distinguish them from holonomies in $\mathbb{C}$. Below is a loose definition of marked periods corresponding to each singularity $Q \in \Sigma$. For a rigorous definition, we refer the reader to Definition 22 of [Ed], which relies on the universal cover of (X,ω) and the lift of a subset of this universal cover into $\mathcal{O}_Q$.

Definition 2.3. Let Q be a singularity of the translation surface (X,ω). A *representative at Q of the marked periods of* (X,ω) is the analogue of holonomy vectors into $\mathcal{O}_Q$ of $\mathrm{SC}_Q(X,\omega)$ in such a way that cumulative angles between saddle connections are preserved between the corresponding marked periods in $\mathcal{O}_Q$. We denote this set of points by $\mathrm{MP}_Q(X,\omega)$.

The term 'representative' is present to emphasize some ambiguity of choice in the definition of marked periods. If Q is of cone angle $2\pi k$, then there are k distinct representatives $\mathrm{MP}_Q(X,\omega)$, each of which is completely determined by the placement of a single marked period into $\mathcal{O}_Q$ of a saddle connection γ beginning at Q. Since $\mathcal{O}_Q$ is comprised of k copies of $\mathbb{C}$, there are k natural choices to place the analogue of $\mathrm{hol}(\gamma)$ into $\mathcal{O}_Q$. Once a choice is made, then the requirement that angles be preserved completely determines the placement into a specific copy of $\mathbb{C}$ of the remaining marked periods: beginning at Q in (X,ω) in the direction of γ, one may sweep counterclockwise through subsequent saddle connections $\delta \in \mathrm{SC}_Q(X,\omega)$ and place the corresponding analogues of $\mathrm{hol}(\delta)$ into $\mathcal{O}_Q$ as the k copies of $\mathbb{C}$ are traversed counterclockwise. After sweeping through the cumulative angle of $2\pi k$ at Q to return to γ, one will have simultaneously swept through all k copies of $\mathbb{C}$ to return to the marked period corresponding to γ in $\mathcal{O}_Q$.

The top of Figure 9 shows (a finite subset of) a representative at the sole singularity Q of the marked periods of the surface (R,ψ). Beginning with the saddle connection that starts at the vertex on the middle of the left edge of (R,ψ) with slope zero, we sweep out an angle of 2π (represented by the single-lined sectors about the vertices) and record marked periods in the first copy of the plane (represented by the single points). We continue to sweep through the second and third cumulative angles of 2π about Q (represented by the double- and triple-lined sectors about the vertices, respectively) and record marked periods in the second and third copies of the plane (represented by the circled and double-circled points, respectively). Had we recorded the marked period corresponding to the beginning saddle connection in the second or third copy of the plane rather than the first, we would obtain another representative $\mathrm{MP}_Q(R,\psi)$ that differs from the one shown by a cyclic permutation of the copies of the plane.

For each marked period $\xi \in \mathrm{MP}_Q(X,\omega)$, one may define the *radius* of ξ to be $d(0,\xi)$ in $\mathcal{O}_Q$, where d is the natural metric inherited from the Euclidean metric on $\mathbb{C}$ (see section 2.5 of [Ed]). For computational purposes, it is convenient to consider only those marked periods bounded by a specified radius $r > 0$.

Definition 2.4. Let $\mathrm{MP}_Q(X,\omega)$ be a representative at Q of the marked periods of a translation surface (X,ω). For $r > 0$, a *radius r bounded representative at Q of the marked periods of* (X,ω) is the set

$$\mathrm{MP}_Q^r(X,\omega) := \{\xi \in \mathrm{MP}_Q(X,\omega) \mid d(0,\xi) \leq r\}.$$

Every saddle connection $\gamma \in \mathrm{SC}(X,\omega)$ is oriented so as to begin at a singularity P and end at a singularity Q. There is an oppositely oriented but otherwise identical saddle connection $-\gamma \in \mathrm{SC}(X,\omega)$ beginning at Q and ending at P. This information is recorded in the set of marked periods corresponding to (X,ω) and is necessary to reconstruct (X,ω) from $\mathcal{O}$.

Definition 2.5 (See Definitions 25 and 26 of [Ed])**.** Let (X,ω) be a translation surface with singular set Σ, and fix an identification $\iota : \sqcup_{Q\in\Sigma}\mathcal{O}_Q \to \mathcal{O}$. For each $Q \in \Sigma$, fix a representative at Q of marked periods $\mathrm{MP}_Q(X,\omega)$. Let μ_2 denote the multiplicative group $\{-1,1\}$. We define a μ_2-action on $\iota(\sqcup_{Q\in\Sigma}\mathrm{MP}_Q(X,\omega)) \subset \mathcal{O}$ where for each $\xi \in \iota(\sqcup_{Q\in\Sigma}\mathrm{MP}_Q(X,\omega))$ corresponding to $\gamma \in \mathrm{SC}(X,\omega)$, the point $-\xi := (-1)\cdot\xi \in \iota(\sqcup_{Q\in\Sigma}\mathrm{MP}_Q(X,\omega))$ corresponds to $-\gamma \in \mathrm{SC}(X,\omega)$. We call the μ_2-space

$$\mathrm{MP}(X,\omega) := \iota(\sqcup_{Q\in\Sigma}\mathrm{MP}_Q(X,\omega)) \subset \mathcal{O}$$

a *representative of the marked periods of* (X,ω).

The top of Figure 9 shows a marked period ξ and its image $-\xi$ under this μ_2-action. Notice that the μ_2-action defined above restricts to the bounded representatives $\{\mathrm{MP}^r_Q(X,\omega)\}_{Q\in\Sigma}$ as $\gamma, -\gamma \in \mathrm{SC}(X,\omega)$ have the same length. Hence we may also define a *radius r bounded representative of the marked periods of* (X,ω),

$$\mathrm{MP}^r(X,\omega) := \iota(\sqcup_{Q\in\Sigma}\mathrm{MP}^r_Q(X,\omega)) \subset \mathcal{O}.$$

As noted above, a representative at Q of the marked periods of (X,ω), $\mathrm{MP}_Q(X,\omega)$, is not canonically defined; nor is a representative of the marked periods of (X,ω), $\mathrm{MP}(X,\omega)$. For each $Q \in \Sigma$ of cone angle $2\pi k$, there are k natural choices of $\mathrm{MP}_Q(X,\omega)$, and each of these choices corresponds to a cyclic permutation of the k copies of the plane comprising $\mathcal{O}_Q$, i.e. to an element of the cyclic permutation group C_k. In a similar way, there are a number of natural choices for an identification of $\sqcup_{Q\in\Sigma}\mathcal{O}_Q$ in $\mathcal{O}$. Each $\mathcal{O}_Q$ with Q of order k must be identified with one of the $n(k)$ copies of $\mathcal{O}_k$ in $\mathcal{O}$, but there are no further restrictions for this choice. Thus if (X,ω) belongs to the stratum $\mathcal{H}(i_1,\ldots,i_s)$, $k_1 < \cdots < k_t$ represent the distinct integers in $i_1,\ldots,i_s$, and $n(k_j)$ denotes the number of times k_j appears in $i_1,\ldots,i_s$, then each identification of $\sqcup_{Q\in\Sigma}\mathcal{O}_Q$ in $\mathcal{O}$ corresponds to an element of

$$S_{n(k_1)} \times \cdots \times S_{n(k_t)},$$

where $S_{n(k_j)}$ is the permutation group on $n(k_j)$ elements. Section 3.4 of [Ed] constructs an equivalence relation so that this ambiguity of choice is removed.

Definition 2.6. Let $[\mathrm{MP}(X,\omega)]$ denote the collection of all choices of representatives of marked periods of (X,ω). Given a representative of marked periods $\mathrm{MP}(X,\omega)$, let

$$\mathcal{P} := \{\sigma : \mathcal{O} \to \mathcal{O} \mid \sigma(\mathrm{MP}(X,\omega)) \in [\mathrm{MP}(X,\omega)]\}.$$

For any μ_2-subspace $S \subseteq \mathrm{MP}(X,\omega)$, let

$$[S] := \{\sigma(S) \mid \sigma \in \mathcal{P}\}.$$

In particular, for $r > 0$, $[\mathrm{MP}^r(X,\omega)]$ denotes the collection of all choices of radius r bounded representatives of marked periods of (X,ω).

Hence the representatives of $[\mathrm{MP}(X,\omega)]$ (or $[\mathrm{MP}^r(X,\omega)]$) may be thought of as differing by permutations of the various copies of $\mathcal{O}_Q \sim \mathcal{O}_k$ for each $k \in \mathbb{N} \cup \{0\}$ and by cyclic permutations of the copies of the plane comprising each individual $\mathcal{O}_Q$ for $Q \in \Sigma$. It is important to recall that $\mathrm{MP}(X,\omega)$ ($\mathrm{MP}^r(X,\omega)$) is a μ_2-space, so the various components of one representative of $[\mathrm{MP}(X,\omega)]$ ($[\mathrm{MP}^r(X,\omega)]$) may be appropriately permuted so that not only do the set of points in $\mathcal{O}$ agree with another representative, but the μ_2-actions agree as well.

Explicit computations for the algorithm are performed using a representative of $[\mathrm{MP}^r(X,\omega)]$ for increasing values r. We introduce a decomposition of a translation surface (X,ω) into 2-cells which gives rise to a specific radius ρ whereby $[(X,\omega)]$ may be reconstructed from a representative $\mathrm{MP}^{2\rho}(X,\omega)$.

Definition 2.7. Let (X,ω) be a translation surface with singular set Σ. For each $Q \in \Sigma$, let

$$V_Q := \{x \in X \mid d(x,Q) \leq d(x,P) \text{ for all } P \in \Sigma\}$$

and

$$\rho_Q := \max_{x \in V_Q}\{d(x,Q)\}.$$

The collection of 2-cells $\{V_Q\}_{Q\in\Sigma}$ is called the *Voronoi decomposition of* (X,ω) *subordinate to* Σ. The *Voronoi radius of* (X,ω) is

$$\rho := \max_{Q\in\Sigma}\{\rho_Q\}.$$

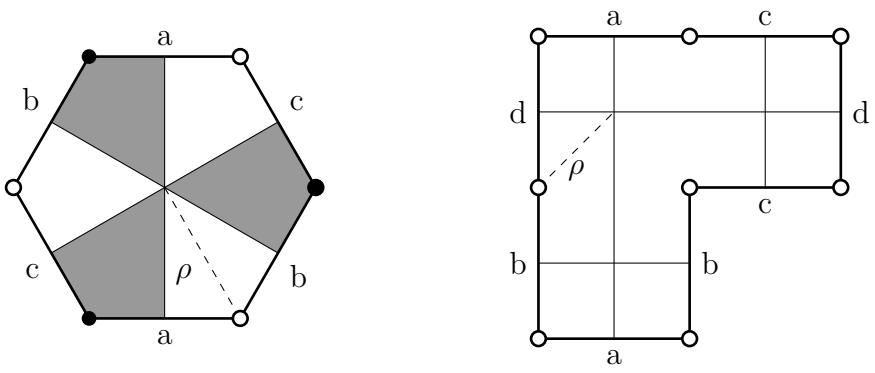


Figure 8: Left: Two distinct 2-cells comprising the Voronoi decomposition of the regular hexagon subordinate to its singular set. Here $\rho = \rho_{Q_j}$ for either singularity Q_j in $\Sigma = \{Q_1, Q_2\}$. Right: A single 2-cell comprising the Voronoi decomposition of (R, ψ). Here $\rho = \sqrt{2}/2$.

Since each V_Q is compact and the distance function is continuous, the maximum distance from Q over points in V_Q is attained by some $x \in V_Q$ and thus ρ_Q exists for each $Q \in \Sigma$. Since Σ is finite, ρ also exists (see Figure 8).

In Section 4.2 of [Ed], Edwards shows that for each singularity $Q \in \Sigma$ of a translation surface (X, ω), the set of points

$$C_Q := \{z \in \mathcal{O}_Q \mid d(z, 0) < d(z, \xi) \text{ for all } \xi \in \mathrm{MP}_Q(X, \omega)\}$$

admits a holomorphic 1-form φ whereby (C_Q, φ) defines a translation surface that is translation equivalent to $(\mathring{V}_Q, \omega|_{\mathring{V}_Q})$, where $\mathring{V}_Q$ denotes the interior of V_Q (Theorem 10 of [Ed]). He then shows in Section 4.3 that the closure of C_Q, which we denote $\overline{C_Q}$, may be constructed from the finite set of marked periods $\mathrm{MP}_Q^{2\rho}(X, \omega)$, and in Section 4.4 Edwards describes how the μ_2-action on $\sqcup_{Q \in \Sigma} \mathrm{MP}_Q(X, \omega)$ provides gluing information for segments of the boundary of $\sqcup_{Q \in \Sigma} \overline{C_Q}$ whereby one may reconstruct (X, ω) up to translation equivalence (this is the right-hand side of Figure 7). This reconstruction leads to the following key result for the Edwards algorithm:

Theorem 2.1 (See Theorem 8 and Corollary 2 of [Ed])**.** Let (X_1, ω_1) and (X_2, ω_2) be two translation surfaces with minimum Voronoi radius ρ. If (X_1, ω_1) and (X_2, ω_2) are translation equivalent, then

$$[\mathrm{MP}(X_1, \omega_1)] = [\mathrm{MP}(X_2, \omega_2)].$$

Conversely, if

$$[\mathrm{MP}^{2\rho}(X_1, \omega_1)] = [\mathrm{MP}^{2\rho}(X_2, \omega_2)],$$

then (X_1, ω_1) and (X_2, ω_2) are translation equivalent.

Edwards defines an $\mathrm{SL}_2\mathbb{R}$-action on the set of marked periods of translation surfaces that corresponds directly to the $\mathrm{SL}_2\mathbb{R}$-action on the set of all translation surfaces themselves. Rather than explicitly define this $\mathrm{SL}_2\mathbb{R}$-action here, we refer the reader to Definition 36 of [Ed] and provide an illustration of the action on the surface (R, ψ).

Remark. In what follows, we abuse notation and identify $[\mathrm{MP}(X, \omega)]$ with a representative $\mathrm{MP}(X, \omega)$. For instance, we may speak of a marked period ξ in $[\mathrm{MP}(X, \omega)]$, by which we mean the orbit of the marked period $\xi \in \mathrm{MP}(X, \omega)$ under the collection of maps $\mathcal{P}$ in Definition 2.6.

Let $N = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$. The lower half of Figure 9 shows a subset of $N \cdot [\mathrm{MP}(R, \psi)]$. This action almost behaves as usual matrix multiplication restricted to each copy of the plane, but some care must be exercised due to the cyclic gluing of the copies of the plane. Consider, for instance, the marked period $\eta \in [\mathrm{MP}(R, \psi)]$ at the point $(2, -1)$ in the first copy of the plane at the top of Figure 9. Usual matrix multiplication gives

$$\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

As the lower slit of the non-negative real axis of the first copy of the plane is identified with the upper slit of the non-negative real axis of the second copy of the plane, the point η 'slides under' the slit in the first copy of the plane and into the second copy under the action of N.

For a general translation surface (X, ω) and matrix $A \in \mathrm{SL}_2\mathbb{R}$, the marked periods of $A \cdot [\mathrm{MP}(X, \omega)]$ inherit the μ_2-action of the marked periods of $[\mathrm{MP}(X, \omega)]$, i.e.

$$-(A \cdot \xi) := A \cdot (-\xi)$$

for all $\xi \in [\mathrm{MP}(X, \omega)]$. For example, consider the marked period ζ at the point $(1, 1)$ in the first copy of the plane of $N \cdot [\mathrm{MP}(R, \psi)]$. This marked period is the image under the action of N of the marked period $\hat{\zeta} \in [\mathrm{MP}(R, \psi)]$ at the point $(1, 0)$ in the first copy of the plane. The marked period $\hat{\zeta}$ corresponds to the saddle connection with slope 0 beginning at the vertex in the middle of the left side of (R, ψ). The identical but oppositely oriented saddle connection

gives rise to the marked period $-\hat{\zeta}$ at the point $(-1, 0)$ in the third copy of the plane. Then $-\zeta := N \cdot (-\hat{\zeta})$. Hence $-\zeta$ is the marked period at the point $(-1, -1)$ in the third copy of the plane of $N \cdot [\mathrm{MP}(R, \psi)]$.

The action of $\mathrm{SL}_2\mathbb{R}$ on the set of marked periods of translation surfaces restricts to give an action on the set of equivalence classes of μ_2-subspaces of representatives of marked periods. In particular, we wish to consider the action of a matrix $A \in \mathrm{SL}_2\mathbb{R}$ on bounded marked periods $[\mathrm{MP}^r(X, \omega)]$ for $r > 0$. Let

$$A \cdot [\mathrm{MP}^r(X, \omega)] := \{\xi \in A \cdot [\mathrm{MP}(X, \omega)] \mid A^{-1} \cdot \xi \in [\mathrm{MP}^r(X, \omega)]\}$$

with μ_2-action inherited from $A \cdot [\mathrm{MP}(X, \omega)]$. Since for all $\xi \in A \cdot [\mathrm{MP}^r(X, \omega)]$,

$$A^{-1} \cdot (-\xi) = -(A^{-1} \cdot \xi) \in [\mathrm{MP}^r(X, \omega)],$$

we see that $-\xi \in A \cdot [\mathrm{MP}^r(X, \omega)]$ and thus $A \cdot [\mathrm{MP}^r(X, \omega)]$ is also a μ_2-space.

The $\mathrm{SL}_2\mathbb{R}$-action defined on the set of marked periods is designed so as to correspond precisely with the $\mathrm{SL}_2\mathbb{R}$-action on the set of translation surfaces:

Lemma 2.1 (See Theorem 16 of [Ed])**.** *For each translation surface (X, ω) and matrix $A \in SL_2\mathbb{R}$,*

$$A \cdot [MP(X, \omega)] = [MP(A \cdot (X, \omega))].$$

This leads us to another characterization of the Veech group, but now in the setting of the model surface $\mathcal{O}$. We present here the proof given by Edwards, as it concisely highlights the various notions introduced thus far and illustrates how they relate to one another.

Theorem 2.2 (See Theorem 17 of [Ed])**.** Given a translation surface (X, ω),

$$\mathrm{SL}(X, \omega) = \mathrm{Stab}_{\mathrm{SL}_2\mathbb{R}}[\mathrm{MP}(X, \omega)].$$

Proof. A matrix A belongs to $\mathrm{SL}(X, \omega)$ if and only if

$$A \cdot [(X, \omega)] = [(X, \omega)]$$

by Definition 1.10. By definition of the action of $\mathrm{SL}_2\mathbb{R}$ on the set of translation equivalent translation surfaces, this is true if and only if

$$[A \cdot (X, \omega)] = [(X, \omega)],$$

which means that $A \cdot (X, \omega)$ and (X, ω) are translation equivalent. By Theorem 2.1, this is true if and only if

$$[\mathrm{MP}(A \cdot (X, \omega))] = [\mathrm{MP}(X, \omega)],$$

which, by Lemma 2.1, is the same as

$$A \cdot [\mathrm{MP}(X, \omega)] = [\mathrm{MP}(X, \omega)].$$

This final equality is precisely the statement that

$$A \in \mathrm{Stab}_{\mathrm{SL}_2\mathbb{R}}[\mathrm{MP}(X, \omega)].$$

□

At this point we have a characterization of the Veech group in terms of the model surface $\mathcal{O}$ by Theorem 2.2, and we know how to represent a translation surface in $\mathcal{O}$ with finite data by Theorem 2.1. The natural question now is how to combine these two theorems so that one may compute the Veech group of (X, ω) using only finite data in $\mathcal{O}$. One might initially conjecture that $\mathrm{SL}(X, \omega)$ is given by

$$\mathrm{Stab}_{\mathrm{SL}_2\mathbb{R}}[\mathrm{MP}^{2\rho}(X, \omega)].$$

However, while $[\mathrm{MP}^{2\rho}(X, \omega)]$ is enough data to reconstruct (X, ω) from $\mathcal{O}$, we generally need to start with a larger subset of $[\mathrm{MP}(X, \omega)]$ in order to meaningfully compare the image of this subset under the action of $A \in \mathrm{SL}_2\mathbb{R}$ with $[\mathrm{MP}^{2\rho}(X, \omega)]$. In Figure 9, we wish to compare the marked periods of the shaded region on the top with those in the shaded region on the bottom. However, the marked periods of $N \cdot [\mathrm{MP}^{2\rho}(R, \psi)]$ are contained inside the dotted ellipses that are the images of the dotted circles of radius 2ρ under N. The marked periods at the points $(-1, 1)$ and $(1, -1)$ in each copy of the plane of $N \cdot [\mathrm{MP}(R, \psi)]$ are the images under N of marked periods in $[\mathrm{MP}(R, \psi)]$ at points $(-1, 2)$ and $(1, -2)$, respectively—points that lie outside of the bounded set $[\mathrm{MP}^{2\rho}(R, \psi)]$.

Thus in general we must begin with a larger subset of $[\mathrm{MP}(X, \omega)]$ in order to compare the image of this subset under the action of A with the set $[\mathrm{MP}^{2\rho}(X, \omega)]$ that determines $[(X, \omega)]$. As seen in the above example, the set of points of $N \cdot [\mathrm{MP}^{2\rho}(R, \psi)]$ in each copy of the plane are contained inside the ellipse that is the image of the circle of radius 2ρ under N. This

phenomenon is not unique to $(R, \psi), N$ nor the radius 2ρ; for any $r > 0$, the set of points of $A \cdot [\mathrm{MP}^r(X, \omega)]$ in each copy of the plane is contained inside the ellipse that is the image of the circle of radius r under A. Thus if we begin with a sufficiently large radius r so that the length of the minor semi-axis of this ellipse is greater than or equal to 2ρ, then we may compare the subset of $A \cdot [\mathrm{MP}^r(X, \omega)]$ that is bounded by 2ρ with the set $[\mathrm{MP}^{2\rho}(X, \omega)]$ that uniquely determines $[(X, \omega)]$. If these μ_2-spaces agree, then $A \in \mathrm{SL}(X, \omega)$.

Each matrix $A \in \mathrm{SL}_2\mathbb{R}$ has a singular value decomposition

$$A = UDV^t$$

where U and V are unitary and D is diagonal with non-negative real entries. The diagonal entries of D, say d and d' with $d \leq d'$, are called the *singular values* of A (not to be confused with the singular set Σ of a translation surface), and these singular values are uniquely determined up to order by A. The singular values d and d' of A can be interpreted geometrically as the lengths of the semi-axes of the ellipse that is the image of the unit circle under A. Since the determinant is multiplicative, $A \in \mathrm{SL}_2\mathbb{R}$, and U and V are unitary, we find

$$1 = \det(A) = \det(U)\det(D)\det(V^t) = \det(D) = dd'.$$

Hence the singular values of $A \in \mathrm{SL}_2\mathbb{R}$ are d, d^{-1} with $d \leq d^{-1}$. Since the Frobenius norm of a matrix A can be written $||A|| = \sqrt{\mathrm{tr}(AA^t)}$ (see the comment below Definition 1.11) and the trace function is invariant under cyclic permutations, we find that

$$\begin{aligned} ||A|| &= \sqrt{\mathrm{tr}((UDV^t)(UDV^t)^t)} \\ &= \sqrt{\mathrm{tr}(UDV^tVD^tU^t)} \\ &= \sqrt{\mathrm{tr}(DD^t)} \\ &= \sqrt{d^2 + d^{-2}}. \end{aligned}$$

Solving for d gives

$$d = \sqrt{\frac{1}{2}\left(||A||^2 - \sqrt{||A||^4 - 4}\right)}.$$

This proves the following useful result:

Lemma 2.2 (See Section 5.3 of [Ed]). *The strictly decreasing function*

$$SV : [\sqrt{2}, \infty) \to (0, 1]$$
$$\mathfrak{b} \mapsto \sqrt{\frac{1}{2}(\mathfrak{b}^2 - \sqrt{\mathfrak{b}^4 - 4})}$$

gives the minimum singular value d of a matrix $A \in SL_2\mathbb{R}$ with Frobenius norm $||A|| = \mathfrak{b}$. That is, $SV(||A||) = d$. Furthermore, SV is invertible with $SV^{-1}(d) = \sqrt{d^2 + d^{-2}}$. Thus for a matrix $A \in SL_2\mathbb{R}$ with minimum singular value d,

$$||A|| = SV^{-1}(d) = \sqrt{d^2 + d^{-2}}.$$

We introduce some notation before stating the main result of Edwards that ties together Theorems 2.1 and 2.2 and provides a way of finding increasing subsets of $\mathrm{SL}(X, \omega)$. Recall Definition 2.6.

Definition 2.8. Let S be a μ_2-subspace of a representative of marked periods $\mathrm{MP}(X, \omega)$. For $r > 0$, define

$$S^r := S \cap \mathrm{MP}^r(X, \omega),$$

and let

$$[S]^r := [S^r].$$

The following theorem allows us to find all elements of the Veech group up to a specified Frobenius norm:

Theorem 2.3 (See Theorem 18 of [Ed]). Let (X, ω) be a translation surface with Voronoi radius ρ. Fix $\mathfrak{b} \geq \sqrt{2}$, and let $\mathfrak{R} = \frac{2\rho}{SV(\mathfrak{b})}$. Then for any $A \in \mathrm{SL}_2\mathbb{R}$ with $||A|| \leq \mathfrak{b}$,

$$A \in \mathrm{SL}(X, \omega)$$

if and only if

$$(A \cdot [\mathrm{MP}^{\mathfrak{R}}(X, \omega)])^{2\rho} = [\mathrm{MP}^{2\rho}(X, \omega)].$$

Given any $\mathfrak{b} \geq \sqrt{2}$, Theorem 2.3 gives an explicit radius $\mathfrak{R}$ such that for any $A \in \mathrm{SL}_2\mathbb{R}$ with $||A|| \leq \mathfrak{b}$, the length ℓ of the minor semi-axis of the ellipse that is the image of the circle of radius $\mathfrak{R}$ under A satisfies $\ell \geq 2\rho$. That is, the set $[\mathrm{MP}^{\mathfrak{R}}(X, \omega)]$ is large enough so that for any matrix A with Frobenius norm bounded above by $\mathfrak{b}$, we may compare the subset of $A \cdot [\mathrm{MP}^{\mathfrak{R}}(X, \omega)]$

that is bounded by 2ρ with the set $[\mathrm{MP}^{2\rho}(X,\omega)]$ that determines $[(X,\omega)]$. These μ_2-sets agree if and only if $A \in \mathrm{SL}(X,\omega)$.

Computationally, Theorem 2.3 gives a way of finding all elements of

$$\mathrm{SL}^{\mathfrak{b}}(X,\omega) := \{A \in \mathrm{SL}(X,\omega) \mid ||A|| \leq \mathfrak{b}\}$$

in finite time (note that this is a finite set due to the discreteness of $\mathrm{SL}(X,\omega)$). Fix two linearly independent elements of the subset of holonomy vectors bounded by 2ρ, say

$$v_0, v_1 \in \mathrm{hol}(\mathrm{SC}(X,\omega)) \cap B(0,2\rho) \subset \mathbb{C}.$$

If $A \in \mathrm{SL}^{\mathfrak{b}}(X,\omega)$, then by Theorem 2.3 (naturally projecting all marked periods into $\mathbb{C}$), A sends some pair of linearly independent vectors

$$w_0, w_1 \in \mathrm{hol}(\mathrm{SC}(X,\omega)) \cap B(0,\mathfrak{R}) \subset \mathbb{C}$$

to v_0 and v_1. Thus if we construct all matrices $A \in \mathrm{SL}_2\mathbb{R}$ sending any of the (finitely many) linearly independent pairs

$$w_0, w_1 \in \mathrm{hol}(\mathrm{SC}(X,\omega)) \cap B(0,\mathfrak{R})$$

to the fixed pair v_0, v_1, then we have a finite set of candidate matrices that necessarily contains $\mathrm{SL}^{\mathfrak{b}}(X,\omega)$. We test each of these candidate matrices A with the criteria of Theorem 2.3: if $||A|| \leq \mathfrak{b}$ and

$$(A \cdot [\mathrm{MP}^{\mathfrak{R}}(X,\omega)])^{2\rho} = [\mathrm{MP}^{2\rho}(X,\omega)],$$

then $A \in \mathrm{SL}^{\mathfrak{b}}(X,\omega)$.

It is important to recall that these sets must be equal as μ_2-spaces. As seen in the comments following Theorem 2.1, the μ_2-action of the representatives of

$$[\mathrm{MP}^{2\rho}(R,\psi)] \qquad \text{and} \qquad (N \cdot [\mathrm{MP}^{\mathfrak{R}}(R,\psi)])^{2\rho}$$

shown in Figure 9 do not agree even though the sets of points do coincide (compare ξ and $-\xi$ with ζ and $-\zeta$). We leave it to the reader to verify that no cyclic permutation of the copies of the planes of these representatives will make the μ_2-actions agree. Hence $N \notin \mathrm{SL}(R,\psi)$.

Now that increasing subsets $\mathrm{SL}^{\mathfrak{b}}(X,\omega) \subseteq \mathrm{SL}(X,\omega)$ may be computed, the question becomes when to halt the algorithm. It is possible that $\mathrm{SL}(X,\omega)$ has no finite set of generators (see [HuSch]), in which case a user-specified iteration limit is reached in the algorithm and the set $\mathrm{SL}^{\mathfrak{b}}(X,\omega)$ is returned for some $\mathfrak{b} \geq \sqrt{2}$. However, if $\mathrm{SL}(X,\omega)$ is detected to be a lattice, then the theory of hyperbolic geometry of Subsection 1.4 allows us to find and return a set of generators of the Veech group.

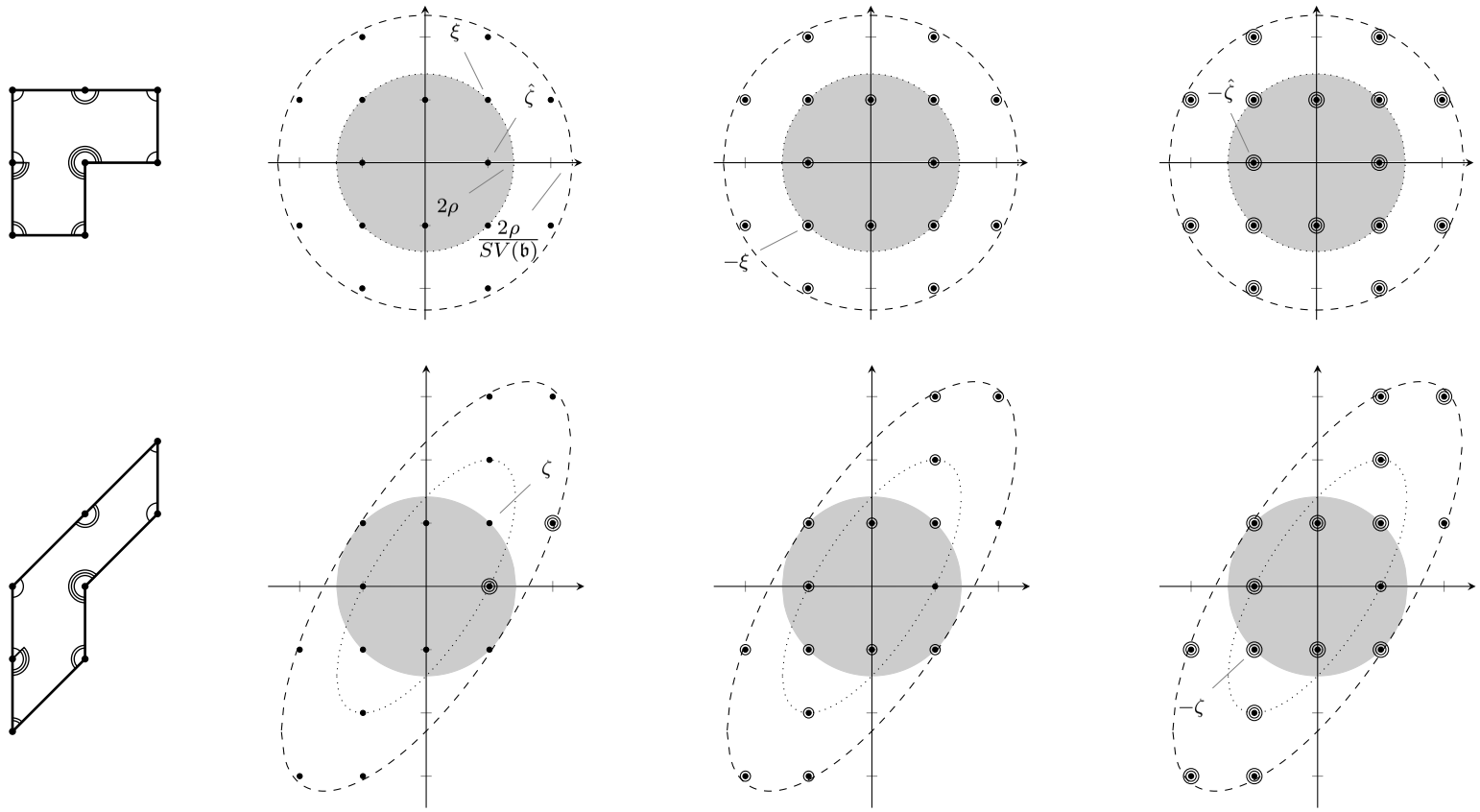


Figure 9: Top: Representatives of the bounded marked periods $[\mathrm{MP}^{2\rho}(R,\psi)]$ (inside dotted/shaded ball) and $[\mathrm{MP}^{\mathfrak{R}}(R,\psi)]$ (inside dashed ball) for $\mathfrak{R} = 2\rho/SV(\mathfrak{b})$ with $\rho = \sqrt{2}/2, \mathfrak{b} \approx 1.7645$. Bottom: Representatives of $N \cdot [\mathrm{MP}^{2\rho}(R,\psi)]$ (inside dotted ellipse), $N \cdot [\mathrm{MP}^{\mathfrak{R}}(R,\psi)]$ (inside dashed ellipse) and $(N \cdot [\mathrm{MP}^{\mathfrak{R}}(R,\psi)])^{2\rho}$ (inside shaded ball) for $N = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$.

2.2 Halting the Algorithm Via Hyperbolic Geometry

Throughout this subsection, we will assume that

$$\mathrm{PSL}(X,\omega) \cap \overline{\mathrm{SO}_2\mathbb{R}} = \{\mathrm{Id}\}$$

(step 2 of Algorithm 2.1 perturbs the original translation surface to ensure that this is the case, and step 11 recovers generators of the Veech group of the original surface by conjugating generators of the Veech group of the perturbed surface). For $\mathfrak{b} \geq \sqrt{2}$, let

$$\mathrm{PSL}^{\mathfrak{b}}(X,\omega) := \overline{\mathrm{SL}^{\mathfrak{b}}(X,\omega)} \subset \mathrm{PSL}_2\mathbb{R}.$$

Lemma 2.3. *Let (X,ω) be a translation surface. The Veech group $SL(X,\omega)$ is a lattice if and only if for some $\mathfrak{b} \geq \sqrt{2}$,*

$$\Omega(PSL^{\mathfrak{b}}(X,\omega))$$

has no free sides.

Proof. If $\mathrm{SL}(X,\omega)$ is a lattice, then the fundamental region $\Omega(\mathrm{PSL}(X,\omega))$ has finite hyperbolic volume, and Lemma 1.4(ii) guarantees that $\Omega(\mathrm{PSL}(X,\omega))$ has finitely many sides and no free sides. The sides of $\Omega(\mathrm{PSL}(X,\omega))$ correspond to the half-planes determined by a finite subset of $\mathrm{PSL}(X,\omega)$, say

$$\{M_1, M_2, \ldots, M_n\} \subset \mathrm{PSL}(X,\omega).$$

Let $\mathfrak{b} = \max\{||M_j||\}_{j=1}^n$ so that $\{M_j\}_{j=1}^n \subseteq \mathrm{PSL}^{\mathfrak{b}}(X,\omega)$. Then

$$\Omega(\mathrm{PSL}^{\mathfrak{b}}(X,\omega)) = \Omega(\mathrm{PSL}(X,\omega)),$$

and so $\Omega(\mathrm{PSL}^{\mathfrak{b}}(X,\omega))$ has no free sides.

Conversely, suppose there is some $\mathfrak{b} \geq \sqrt{2}$ such that $\Omega(\mathrm{PSL}^{\mathfrak{b}}(X,\omega))$ has no free sides. Since $\mathrm{PSL}^{\mathfrak{b}}(X,\omega)$ is finite, $\Omega(\mathrm{PSL}^{\mathfrak{b}}(X,\omega))$ has finitely many sides and Corollary 1.1 guarantees that $\nu(\Omega(\mathrm{PSL}^{\mathfrak{b}}(X,\omega))) < \infty$. But

$$\begin{aligned}
\Omega(\mathrm{PSL}(X,\omega)) &= \bigcap_{M \in \mathrm{PSL}(X,\omega)\setminus\{\mathrm{Id}\}} H_i(M \cdot i) \\
&\subseteq \bigcap_{M \in \mathrm{PSL}^{\mathfrak{b}}(X,\omega)\setminus\{\mathrm{Id}\}} H_i(M \cdot i) \\
&= \Omega(\mathrm{PSL}^{\mathfrak{b}}(X,\omega)),
\end{aligned}$$

and so $\nu(\Omega(\mathrm{PSL}(X,\omega))) < \infty$. Since $\Omega(\mathrm{PSL}(X,\omega))$ is a fundamental region for $\mathrm{SL}(X,\omega)$, we see that $\mathrm{SL}(X,\omega)$ is a lattice. □

If the algorithm detects that $\Omega(\mathrm{PSL}^{\mathfrak{b}}(X,\omega))$ has no free sides (before a user-specified iteration limit is reached), then it will continue to run until generators of the lattice group $\mathrm{SL}(X,\omega)$ are found and returned. We introduce a function that will help to determine when $\mathrm{SL}^{\mathfrak{b}}(X,\omega)$ contains generators of $\mathrm{SL}(X,\omega)$.

Lemma 2.4 (See Definition 50 and Theorem 22 of [Ed])**.** *The function*

$$\begin{aligned}\chi : [\sqrt{2},\infty) &\to [0,\infty)\\ t &\mapsto -\ln(SV(t))\end{aligned}$$

satisfies the following properties:

(i) for $M \in SL_2\mathbb{R}$, $\chi(||M||)$ gives the shortest hyperbolic distance from i to the perpendicular bisector of i and $M \cdot i$, i.e.

$$\chi(||M||) = \frac{1}{2}\rho(i, M\cdot i);$$

(ii) for $M \in SL_2\mathbb{R}$,

$$B(i,\chi(||M||)) \subset H_i(M\cdot i)$$

where $B(i,\chi(||M||))$ is the hyperbolic ball of radius $\chi(||M||)$ centered at i, and;

(iii) χ is strictly increasing.

Proof. We refer the reader to Theorem 22 of [Ed] for a proof of part (i) of Lemma 2.4. Part (ii) is an immediate consequence of (i): if $\tau \in B(i,\chi(||M||))$, then

$$\rho(i,\tau) < \chi(||M||) = \frac{1}{2}\rho(i, M\cdot i),$$

and $H_i(M\cdot i)$ consists of all points in $\mathbb{H}$ that are closer to i than to $M\cdot i$. Part (iii) follows from the fact that $SV : [\sqrt{2},\infty) \to (0,1]$ is strictly decreasing, as is $(-\ln)$. Thus the composition $\chi = -\ln \circ SV$ is strictly increasing. □

It can be shown that hyperbolic balls are in fact Euclidean balls (with different centers). For any $\mathfrak{b} \geq \sqrt{2}$ and $S \subseteq \mathrm{PSL}(X,\omega)$, let

$$Bdd_{\chi(\mathfrak{b})}(S) := \Omega(S) \cap B(i,\chi(\mathfrak{b}))$$

denote the portion of $\Omega(S)$ that is bounded by a hyperbolic distance of $\chi(\mathfrak{b})$ from i. Figure 10 shows this set for $S = \mathrm{PSL}^{\mathfrak{b}}(M_0 \cdot (R, \psi))$ with $M_0 = \left(\begin{smallmatrix} 1 & 0 \\ 1/2 & 1 \end{smallmatrix}\right)$ and $\mathfrak{b} \approx 4.0078$. (The matrix M_0 is introduced so that $\mathrm{PSL}(M_0 \cdot (R, \psi))$ trivially stabilizes i, which is not the case for $\mathrm{PSL}(R, \psi)$.) Lemma 2.4 implies the following:

Lemma 2.5 (See Corollary 3 and Lemma 7 of [Ed]). *For any* $\mathfrak{b} \geq \sqrt{2}$,

$$Bdd_{\chi(\mathfrak{b})}(PSL(X, \omega)) = Bdd_{\chi(\mathfrak{b})}(PSL^{\mathfrak{b}}(X, \omega)).$$

Proof. Any matrix $M \in \mathrm{PSL}(X, \omega) \backslash \mathrm{PSL}^{\mathfrak{b}}(X, \omega)$ satisfies $||M|| > \mathfrak{b}$, and so by parts (ii) and (iii) of Lemma 2.4,

$$B(i, \chi(\mathfrak{b})) \subset B(i, \chi(||M||)) \subset H_i(M \cdot i).$$

Since

$$\Omega(\mathrm{PSL}(X, \omega) \backslash \mathrm{PSL}^{\mathfrak{b}}(X, \omega)) = \bigcap_{M \in \mathrm{PSL}(X,\omega) \backslash \mathrm{PSL}^{\mathfrak{b}}(X,\omega)} H_i(M \cdot i),$$

we find

$$\Omega(\mathrm{PSL}(X, \omega) \backslash \mathrm{PSL}^{\mathfrak{b}}(X, \omega)) \cap B(i, \chi(\mathfrak{b})) = B(i, \chi(\mathfrak{b})).$$

Writing

$$\Omega(\mathrm{PSL}(X, \omega)) = \Omega(\mathrm{PSL}^{\mathfrak{b}}(X, \omega)) \cap \Omega(\mathrm{PSL}(X, \omega) \backslash \mathrm{PSL}^{\mathfrak{b}}(X, \omega)),$$

we have

$$\begin{aligned} Bdd_{\chi(\mathfrak{b})}(\mathrm{PSL}(X, \omega)) &= \Omega(\mathrm{PSL}(X, \omega)) \cap B(i, \chi(\mathfrak{b})) \\ &= \Omega(\mathrm{PSL}^{\mathfrak{b}}(X, \omega)) \cap B(i, \chi(\mathfrak{b})) \\ &= Bdd_{\chi(\mathfrak{b})}(\mathrm{PSL}^{\mathfrak{b}}(X, \omega)). \end{aligned}$$

□

As $\mathfrak{b}$ increases, the sets $\Omega(\mathrm{PSL}^{\mathfrak{b}}(X, \omega))$ nest down toward $\Omega(\mathrm{PSL}(X, \omega))$ while the radius $\chi(\mathfrak{b})$ increases. Thus for increasing $\mathfrak{b}$, the nesting convex sets $\Omega(\mathrm{PSL}^{\mathfrak{b}}(X, \omega))$ give more and more accurate approximations to $\Omega(\mathrm{PSL}(X, \omega))$ while our knowledge of where $\Omega(\mathrm{PSL}^{\mathfrak{b}}(X, \omega))$ and $\Omega(\mathrm{PSL}(X, \omega))$ coincide simultaneously increases. In general, $\Omega(\mathrm{PSL}(X, \omega))$ has vertices at infinity and so each $B(i, \chi(\mathfrak{b}))$ will 'cut off' some parts of the polygons $\Omega(\mathrm{PSL}^{\mathfrak{b}}(X, \omega))$ and $\Omega(\mathrm{PSL}(X, \omega))$ (see Figure 10). The following theorem gives sufficient conditions for a subset of $\mathrm{PSL}^{\mathfrak{b}}(X, \omega)$ to generate $\mathrm{PSL}(X, \omega)$ despite our lack of knowledge outside of $B(i, \chi(\mathfrak{b}))$.

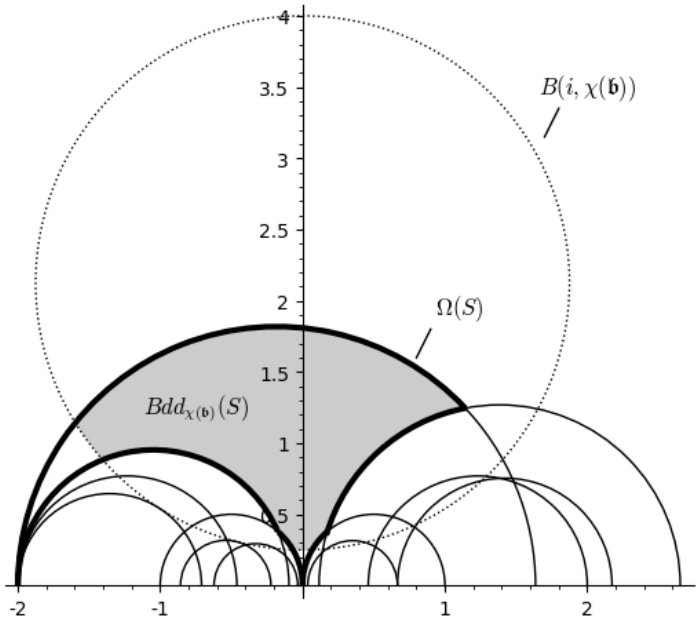


Figure 10: The hyperbolic polygon (bold), bounding ball (dotted) and bounded region (shaded) of $S := \mathrm{PSL}^{\mathfrak{b}}(M_0 \cdot (R, \psi))$, where $M_0 = \left(\begin{smallmatrix} 1 & 0 \\ 1/2 & 1 \end{smallmatrix}\right)$ and $\mathfrak{b} \approx 4.0078$. The remaining h-lines are the boundaries of half-planes in the construction of $\Omega(S)$. (Four h-lines were removed to scale the image.)

Theorem 2.4. Let $\mathfrak{b} \geq \sqrt{2}$ and let $\{M_j\}_{j=1}^n \subset \mathrm{PSL}^{\mathfrak{b}}(X, \omega)$ be the collection of transformations that determine the sides of $\Omega(\mathrm{PSL}^{\mathfrak{b}}(X, \omega))$, i.e. $\{M_j\}_{j=1}^n$ is the minimal subset of $\mathrm{PSL}^{\mathfrak{b}}(X, \omega)$ such that

$$\Omega(\{M_j\}_{j=1}^n) = \Omega(\mathrm{PSL}^{\mathfrak{b}}(X, \omega)).$$

If

$$\nu(Bdd_{\chi(\mathfrak{b})}(\mathrm{PSL}^{\mathfrak{b}}(X, \omega))) > \frac{1}{2}\nu(\Omega(\mathrm{PSL}^{\mathfrak{b}}(X, \omega))),$$

then $\mathrm{PSL}(X, \omega)$ is generated by $\{M_j\}_{j=1}^n$.

Proof. Let $G = \langle M_1, \ldots, M_n \rangle$ be the subgroup of $\mathrm{PSL}(X, \omega)$ generated by $\{M_j\}_{j=1}^n$. We have

$$\Omega(G) \subseteq \Omega(\{M_j\}_{j=1}^n) = \Omega(\mathrm{PSL}^{\mathfrak{b}}(X, \omega)),$$

so

$$\nu(\Omega(G)) \leq \nu(\Omega(\mathrm{PSL}^{\mathfrak{b}}(X,\omega))).$$

Lemma 2.5 together with our hypotheses gives

$$\begin{aligned}\nu(\Omega(\mathrm{PSL}(X,\omega))) &\geq \nu(Bdd_{\chi(\mathfrak{b})}(\mathrm{PSL}(X,\omega))) \\ &= \nu(Bdd_{\chi(\mathfrak{b})}(\mathrm{PSL}^{\mathfrak{b}}(X,\omega))) \\ &> \frac{1}{2}\nu(\Omega(\mathrm{PSL}^{\mathfrak{b}}(X,\omega))) \\ &\geq \frac{1}{2}\nu(\Omega(G)).\end{aligned}$$

By part (i) of Lemma 1.4, $\Omega(G)$ and $\Omega(\mathrm{PSL}(X,\omega))$ are fundamental regions for the groups G and $\mathrm{PSL}(X,\omega)$, respectively, and Lemma 1.3 gives

$$|\mathrm{PSL}(X,\omega) : G| = \frac{\nu(\Omega(G))}{\nu(\Omega(\mathrm{PSL}(X,\omega)))} < 2.$$

Hence $G = \mathrm{PSL}(X,\omega)$. □

2.3 The Algorithm in Pseudocode

Algorithm 2.1. (Compare with Algorithms 7.1 and 7.2 of [Ed])**.**
Input: A polygonally presented translation surface (X_0,ω_0) and an iteration limit $IterationLimit \in \mathbb{N}$.

1. Let $n := 1$, $M_0 := I_2$ and $TrivialStabilizer := FALSE$.
2. While $TrivialStabilizer = FALSE$:
   (i) If $n > 1$, let $M_0 := \left(\begin{smallmatrix} 1 & 0 \\ 1/n & 1 \end{smallmatrix}\right)$.
   (ii) Let $(X,\omega) := M_0 \cdot (X_0,\omega_0)$.
   (iii) Calculate the Voronoi radius ρ of (X,ω).
   (iv) Calculate a representative $\mathrm{MP}^{2\rho}(X,\omega)$.
   (v) Calculate $\mathrm{SL}^{\sqrt{2}}(X,\omega) = \mathrm{SL}(X,\omega) \cap \mathrm{SO}_2\mathbb{R}$ using Theorem 2.3.
   (vi) If $\mathrm{SL}^{\sqrt{2}}(X,\omega) \subseteq \{\pm I_2\}$, let $TrivialStabilizer := TRUE$.
   (vii) Increase n by 1.

3. If $-I_2 \in \mathrm{SL}^{\sqrt{2}}(X,\omega)$, let $ContainsMinusIdentity := TRUE$; else, let $ContainsMinusIdentity := FALSE$.

4. Let $GuaranteedLattice := FALSE$, $Iteration := 1$ and $\mathfrak{R} := 2\rho$.

5. While $GuaranteedLattice = FALSE$ and $Iteration \leq IterationLimit$:

   (i) Double $\mathfrak{R}$ and let $\mathfrak{b} := SV^{-1}(2\rho/\mathfrak{R})$.

   (ii) Calculate a representative $\mathrm{MP}^{\mathfrak{R}}(X,\omega)$.

   (iii) Calculate $\mathrm{SL}^{\mathfrak{b}}(X,\omega)$ using Theorem 2.3.

   (iv) Construct $\Omega(\mathrm{PSL}^{\mathfrak{b}}(X,\omega))$.

   (v) If $\Omega(\mathrm{PSL}^{\mathfrak{b}}(X,\omega))$ has no free sides, let $GuaranteedLattice := TRUE$.

   (vi) Increase *Iteration* by 1.

6. If $GuaranteedLattice = FALSE$, return $\mathrm{SL}^{\mathfrak{b}}(X,\omega)$; else, continue.

7. Let $BoundedRegion := \Omega(\mathrm{PSL}^{\mathfrak{b}}(X,\omega)) \cap B(i,\chi(\mathfrak{b}))$.

8. While $\nu(BoundedRegion) \leq \frac{1}{2}\nu(\Omega(\mathrm{PSL}^{\mathfrak{b}}(X,\omega)))$:

   (i) Double $\mathfrak{R}$ and let $\mathfrak{b} := SV^{-1}(2\rho/\mathfrak{R})$.

   (ii) Calculate a representative $\mathrm{MP}^{\mathfrak{R}}(X,\omega)$.

   (iii) Calculate $\mathrm{SL}^{\mathfrak{b}}(X,\omega)$ using Theorem 2.3.

   (iv) Construct $\Omega(\mathrm{PSL}^{\mathfrak{b}}(X,\omega))$ and let

   $$BoundedRegion := \Omega(\mathrm{PSL}^{\mathfrak{b}}(X,\omega)) \cap B(i,\chi(\mathfrak{b})).$$

9. Let $SideRepresentatives$ be the elements of $\mathrm{PSL}^{\mathfrak{b}}(X,\omega)$ that form the sides of $\Omega(\mathrm{PSL}^{\mathfrak{b}}(X,\omega))$.

10. If $ContainsMinusIdentity = TRUE$, let

    $$AssociatedGenerators := SideRepresentatives \cup \{-I\};$$

    else, let

    $$AssociatedGenerators := SideRepresentatives.$$

11. Let $Generators := M_0^{-1} \cdot AssociatedGenerators \cdot M_0$.

12. Return $Generators$.

2.4 Improvements to the Edwards Algorithm

We briefly mention some improvements that have been made to the original algorithm in [Ed]. Whereas Edwards presents two algorithms—one for the case that $\mathrm{PSL}(X,\omega)$ trivially stabilizes i and the second for the more general case, which calls on the first—Algorithm 2.1 streamlines these two algorithms to present them in tandem. Step 5 of Algorithm 7.2 of Edwards requires finding some $M_0 \in \mathrm{SL}_2\mathbb{R}$ such that $\mathrm{SL}(M_0 \cdot (X,\omega)) \cap \mathrm{SO}_2\mathbb{R} \subseteq \{\pm I_2\}$. Step 2 of Algorithm 2.1 provides explicit candidates for M_0. The following lemma guarantees that this loop is exited in finite time:

Lemma 2.6. *Given a translation surface (X,ω) with*

$$SL(X,\omega) \cap SO_2\mathbb{R} \not\subseteq \{\pm I_2\},$$

there exists some $n \in \mathbb{N}$ such that

$$SL(M_n \cdot (X,\omega)) \cap SO_2\mathbb{R} \subseteq \{\pm I_2\},$$

where $M_n = \left(\begin{smallmatrix} 1 & 0 \\ 1/n & 1 \end{smallmatrix}\right)$.

Proof. Suppose for the sake of contradiction that no such n exists. Then for any $n \in \mathbb{N}$ there is some $R_n \notin \{\pm I_2\}$ such that

$$R_n \in \mathrm{SL}(M_n \cdot (X,\omega)) \cap \mathrm{SO}_2\mathbb{R}.$$

A quick argument using Definition 1.10 of the Veech group reveals that

$$\mathrm{SL}(M_n \cdot (X,\omega)) = M_n \cdot \mathrm{SL}(X,\omega) \cdot M_n^{-1}$$

(this fact is used more generally in step 11 of Algorithm 2.1), and thus

$$R_n \in (M_n \cdot \mathrm{SL}(X,\omega) \cdot M_n^{-1}) \cap \mathrm{SO}_2\mathbb{R}.$$

Conjugating both sides by M_n^{-1} gives $M_n^{-1} R_n M_n \in S_n$ where

$$S_n := \mathrm{SL}(X,\omega) \cap (M_n^{-1} \cdot \mathrm{SO}_2\mathbb{R} \cdot M_n).$$

Now $R_n \in \mathrm{SO}_2\mathbb{R} \backslash \{\pm I_2\}$ is of the form

$$\begin{pmatrix} \cos\theta_n & -\sin\theta_n \\ \sin\theta_n & \cos\theta_n \end{pmatrix}$$

for some $\theta_n \in (0,\pi) \cup (\pi, 2\pi)$, and so

$$M_n^{-1} R_n M_n = \begin{pmatrix} \cos\theta_n - \frac{1}{n}\sin\theta_n & -\sin\theta_n \\ (1+\frac{1}{n^2})\sin\theta_n & \cos\theta_n + \frac{1}{n}\sin\theta_n \end{pmatrix}. \tag{2}$$

Notice that the matrices $M_n^{-1} R_n M_n$ are distinct for various $n \in \mathbb{N}$: if $n \neq m$ and

$$M_n^{-1} R_n M_n = M_m^{-1} R_m M_m,$$

then

$$\sin\theta_n = \sin\theta_m, \qquad \left(1 + 1/n^2\right)\sin\theta_n = (1 + 1/m^2)\sin\theta_m$$

imply $\sin\theta_n = 0$ and we find the contradiction that $\theta_n \in \{0, \pi\}$. Hence $S = \cup_{n\in\mathbb{N}} S_n$ is an infinite set. A general element $A_n \in S_n$ is of the form (2) with $\theta_n \in [0, 2\pi)$. One finds with a quick calculation that the Frobenius norm (Definition 1.11) of A_n is

$$||A_n|| = \sqrt{2 + \left(\frac{4}{n^2} + \frac{1}{n^4}\right)\sin^2\theta_n} \leq \sqrt{7}.$$

As this upper bound is independent of n, we see that S is contained in a closed ball of radius $\sqrt{7}$, i.e. a compact set. But $S \subseteq \mathrm{SL}(X,\omega)$ contains infinitely many elements, so it contains a convergent sequence. This contradicts the discreteness of $\mathrm{SL}(X,\omega)$ (see comments following Definition 1.12). □

The calculation of the volume of the hyperbolic polygon in step 8(e) of Algorithm 7.1 of [Ed] has been replaced with the more computationally efficient check of free sides in step 5(v) of Algorithm 2.1. These steps are equivalent since a hyperbolic polygon with finitely many sides has finite hyperbolic volume if and only if it has no free sides—if such a polygon $\mathcal{P}$ has a free side, then it has infinite volume as seen from the integral definition of $\nu(\mathcal{P})$; the reverse implication follows from Corollary 1.1. The most notable improvement to the algorithm is that only one of the three conditions of step 13 of Algorithm 7.1 of [Ed] is checked in step 8 of Algorithm 2.1. The two extraneous conditions were originally given to ensure that the hyperbolic polygon $\Omega(\mathrm{PSL}^{\mathfrak{b}}(X,\omega))$ is in fact a fundamental region for the subgroup $G \leq \mathrm{PSL}(X,\omega)$ that is generated by the subset of $\mathrm{PSL}^{\mathfrak{b}}(X,\omega)$ determining the sides of $\Omega(\mathrm{PSL}^{\mathfrak{b}}(X,\omega))$. Theorem 2.4 shows that the volume condition alone is strong enough to guarantee that $G = \mathrm{PSL}(X,\omega)$.

3 Appendix: Successfully Computed Veech Groups and Access to the Algorithm

Tables 1,2 and 3 present a collection of translation surfaces and generators of their Veech groups which the algorithm has successfully computed. The name of each translation surface as called in the sage-flatsurf package of SageMath is provided for reference ([DeHo],[Sa]). The 'Verified' column highlights which Veech groups the author has compared with other sources. To access the algorithm, please contact the author at sandesla@oregonstate.edu.

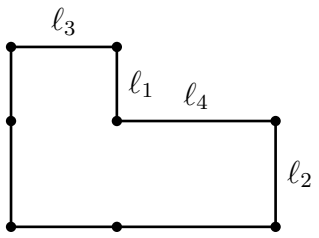


Figure 11: The translation surface $L(\ell_1, \ell_2, \ell_3, \ell_4)$, where the various ℓ_j's are the lengths of the corresponding sides. Note that $L(1,1,1,1)$ is translation equivalent to (R, ψ) from Section 2.

Translation Surface	Veech Group Generators	Verified
$L(1,1,1,1)$: mcmullen_L(1,1,1,1)	$\begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$	✓
$L(2,1,1,2)$: mcmullen_L(2,1,1,2)	$\begin{pmatrix} 3 & 1 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}, \begin{pmatrix} 6 & 5 \\ -5 & -4 \end{pmatrix}, \begin{pmatrix} 7 & 4 \\ -9 & -5 \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$	
$L(1,1,1,2)$: mcmullen_L(1,1,1,2)	$\begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}, \begin{pmatrix} 1 & -3 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 3 & -4 \\ -2 & 3 \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$	
$L(2,1,2,2)$: mcmullen_L(2,1,2,2)	$\begin{pmatrix} 1 & 0 \\ -\frac{3}{2} & 1 \end{pmatrix}, \begin{pmatrix} 1 & -4 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 3 & -4 \\ -2 & 3 \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$	

Table 1: The Veech groups of various L-shaped surfaces; see Figure 11.

Translation Surface	Veech Group Generators	a (if applicable)	Verified
Square torus: square_torus() or veech_2n_gon(2)	$\begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$		✓
Regular hexagon: veech_2n_gon(3)	$\begin{pmatrix} \frac{5}{2} & \frac{3}{2}a \\ -\frac{1}{2}a & -\frac{1}{2} \end{pmatrix}, \begin{pmatrix} -\frac{1}{2} & \frac{1}{2}a \\ -\frac{1}{2}a & -\frac{1}{2} \end{pmatrix}, \begin{pmatrix} \frac{5}{2} & \frac{1}{2}a \\ -\frac{3}{2}a & -\frac{1}{2} \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$	$a = \sqrt{3}$	✓
Regular octagon: regular_octagon() or veech_2n_gon(4)	$\begin{pmatrix} \frac{1}{2}a & -\frac{1}{2}a \\ \frac{1}{2}a & \frac{1}{2}a \end{pmatrix}, \begin{pmatrix} a+2 & a+1 \\ -a-1 & -a \end{pmatrix}, \begin{pmatrix} \frac{1}{2}a+2 & \frac{1}{2}a \\ -\frac{3}{2}a-2 & -\frac{1}{2}a \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$	$a = \sqrt{2}$	✓
Regular decagon: veech_2n_gon(5)	$\begin{pmatrix} \frac{1}{2}a^2 - \frac{3}{2} & \frac{1}{2}a \\ -\frac{1}{2}a & \frac{1}{2}a^2 - \frac{3}{2} \end{pmatrix}, \begin{pmatrix} -\frac{3}{2}a^2 + 6 & -\frac{3}{2}a^3 + \frac{11}{2}a \\ \frac{1}{2}a^3 - \frac{5}{2}a & \frac{3}{2}a^2 - 4 \end{pmatrix}, \begin{pmatrix} -\frac{3}{2}a^2 + 6 & -\frac{1}{2}a^3 + \frac{5}{2}a \\ \frac{3}{2}a^3 - \frac{11}{2}a & \frac{3}{2}a^2 - 4 \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$	$a = \sqrt{\frac{5-\sqrt{5}}{2}}$	✓
Regular dodecagon: veech_2n_gon(6)	$\begin{pmatrix} -\frac{1}{2}a & \frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2}a \end{pmatrix}, \begin{pmatrix} a+3 & a+2 \\ -a-2 & -a-1 \end{pmatrix}, \begin{pmatrix} -a-\frac{1}{2} & -\frac{1}{2}a-1 \\ \frac{3}{2}a+3 & a+\frac{5}{2} \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$	$a = \sqrt{3}$	✓

Table 2: The Veech groups of various regular polygons.

Translation Surface	Veech Group Generators	a (if applicable)	Verified
Double pentagon: veech_double_n_gon(5)	$\left(\begin{array}{cc} -\frac{1}{2}a^2+1 & \frac{1}{2}a^3-\frac{3}{2}a \\ -\frac{1}{2}a^3+\frac{3}{2}a & -\frac{1}{2}a^2+1 \end{array}\right), \left(\begin{array}{cc} \frac{1}{2}a^2+\frac{1}{2} & \frac{2}{5}a^3-\frac{1}{2}a \\ -\frac{1}{2}a & -\frac{1}{2}a^2+\frac{3}{2} \end{array}\right), \left(\begin{array}{cc} -\frac{1}{2}a^2+\frac{1}{2} & -\frac{3}{5}a^3+\frac{3}{2}a \\ a^3-\frac{5}{2}a & \frac{1}{2}a^2-\frac{1}{2} \end{array}\right), \left(\begin{array}{cc} -1 & 0 \\ 0 & -1 \end{array}\right)$	$a=\sqrt{\frac{5+\sqrt{5}}{2}}$	✓
Hexagon, two triangles: ward(3)	$\left(\begin{array}{cc} \frac{3}{2} & \frac{1}{6}a \\ -\frac{1}{2}a & \frac{1}{2} \end{array}\right), \left(\begin{array}{cc} \frac{1}{2} & \frac{5}{6}a \\ -\frac{1}{2}a & -\frac{1}{2} \end{array}\right), \left(\begin{array}{cc} \frac{1}{2} & \frac{1}{2}a \\ -\frac{1}{2}a & \frac{1}{2} \end{array}\right), \left(\begin{array}{cc} -1 & 0 \\ 0 & -1 \end{array}\right)$	$a=\sqrt{3}$	✓
Octagon, two squares: ward(4)	$\left(\begin{array}{cc} -\frac{1}{2}a & \frac{1}{2}a \\ -\frac{1}{2}a & -\frac{1}{2}a \end{array}\right), \left(\begin{array}{cc} \frac{1}{2}a+2 & \frac{1}{2}a+1 \\ -\frac{1}{2}a-1 & -\frac{1}{2}a \end{array}\right), \left(\begin{array}{cc} \frac{1}{2}a+1 & \frac{1}{2}a \\ -\frac{3}{2}a-1 & -\frac{1}{2}a \end{array}\right), \left(\begin{array}{cc} -1 & 0 \\ 0 & -1 \end{array}\right)$	$a=\sqrt{2}$	✓
Decagon, two pentagons: ward(5)	$\left(\begin{array}{cc} \frac{1}{2}a^2-\frac{3}{2} & \frac{1}{2}a \\ -\frac{1}{2}a & \frac{1}{2}a^2-\frac{3}{2} \end{array}\right), \left(\begin{array}{cc} -a^2+\frac{9}{2} & -\frac{11}{10}a^3+4a \\ \frac{1}{2}a^3-2a & a^2-\frac{5}{2} \end{array}\right), \left(\begin{array}{cc} -a^2+4 & -\frac{2}{5}a^3+2a \\ a^3-4a & a^2-3 \end{array}\right), \left(\begin{array}{cc} -1 & 0 \\ 0 & -1 \end{array}\right)$	$a=\sqrt{\frac{5-\sqrt{5}}{2}}$	✓
Origami((1,2,3),(1,4)): origami((1,2,3),(1,4))	$\left(\begin{array}{cc} 1 & 0 \\ -2 & 1 \end{array}\right), \left(\begin{array}{cc} 1 & -3 \\ 0 & 1 \end{array}\right), \left(\begin{array}{cc} 3 & -4 \\ -2 & 3 \end{array}\right), \left(\begin{array}{cc} -1 & 0 \\ 0 & -1 \end{array}\right)$		✓
Origami((1,4),(1,2,3)): origami((1,4),(1,2,3))	$\left(\begin{array}{cc} 1 & 0 \\ -3 & 1 \end{array}\right), \left(\begin{array}{cc} 1 & -2 \\ 0 & 1 \end{array}\right), \left(\begin{array}{cc} 3 & -2 \\ -4 & 3 \end{array}\right), \left(\begin{array}{cc} -1 & 0 \\ 0 & -1 \end{array}\right)$		✓

Table 3: The Veech groups of various surfaces comprised of multiple polygons. Origamis are defined by two permutations of the labels of unit squares; the first permutation specifies how squares are glued from left to right, and the second how they are glued from bottom to top.

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